

# A One-Loop BV Obstruction in Pure Weyl Gravity and Its Formal $\tau$ -adic Wess–Zumino Resolution

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## Abstract

We compute the coefficient-bearing one-loop BV anomaly of strict four-dimensional pure Weyl gravity in the complete gauge-fixed local quotient on the declared regular Bach-locus chart. The repository Euclidean elliptic complex, measure, zero-mode, contour, parity, and local heat-kernel ledgers give

$$\mathcal{A}^{(1)} = \frac{1}{(4\pi)^2} \left( \frac{199}{30} [\omega C^2] - \frac{87}{20} [\omega E_4] \right).$$

Both coordinates are nonzero in the strict quotient. Hence the fixed-field-content local Euclidean quantum master equation is obstructed at one loop. An exact dual-cone witness further excludes cancellation by any nonnegative combination of standard-sign free conformal scalars, Weyl or Dirac fermions, and gauge vectors.

We then adjoin a shifting scalar  $\tau$  and derive its complete minimal-BV cotangent lift from  $S_{\tau, \min} = \int \tau^* (\mathcal{L}_\xi \tau + \omega)$ . In dressed canonical variables

$$\hat{g} = e^{-2\tau} g, \quad \hat{g}^* = e^{2\tau} g^*, \quad \hat{\tau}^* = \tau^* + 2g_{\mu\nu} g^{*\mu\nu},$$

the Weyl sector is an exact contractible quartet. In the formal  $\tau$ -adic local analytic jet algebra the complete dimension-four extended quotient is

$$\dim H_{\text{ext}}^{0,4} = (3 \text{ even}, 1 \text{ odd}), \quad H_{\text{ext}}^{1,4} = 0.$$

The coefficient-bearing Wess–Zumino counterterm therefore restores the extended local Euclidean QME at one loop.

The two dispositions refer to different theories and are not contradictory: strict field content is obstructed, while the formal  $\tau$ -adic compensator extension is locally restored at one loop. Under an explicitly declared local quantum action principle, the complete extended  $H^{1,4} = 0$  result and stable four-direction  $H^{0,4}$  module give a conditional order-by-order formal local QME restoration theorem. No all-order regulator implementing that hypothesis, Lorentzian QME, or equivalence to the strict theory is claimed.

Separately, an exact synthesis now exhausts the declared minimal classical compensator ladder: passive  $\tau$ , minimal polar/HT, quadratic  $P(X)$ , first polynomial braiding, literal and convention-correct linear- $F$  curvature couplings, and the minimal real Weyl connection. Each declared component has empty good locus under its own hypotheses; this is not closure under hybrids or a universal scalar–tensor no-go. No classical action is selected for a determinant or QAP freeze. A separated real scale and compact internal  $U(1)$  connection is only the next preflight, not a selected theory.

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OpenAI GPT-5 was used substantively for mathematical synthesis, verification code, claim auditing, and manuscript drafting. The human author directed the programme and remains responsible for the claims, citations, and final text. All cohomological ranks and cancellations reported here use exact rational arithmetic. Every headline statement is bound to a deterministic certificate and an independent verifier. Submission remains conditional on final human author review.

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## 1 Question, result, and claim boundary

Pure Weyl gravity has classical  $\text{Diff} \times \text{Weyl}$  gauge symmetry. The quantum question is not whether a determinant has a nonzero logarithm, but whether the regularized Slavnov breaking is trivial in the relative BV cohomology  $H^{1,4}(s \mid d)$ . We therefore separate three calculations:

1. classify the complete local obstruction space;
2. compute the coefficient vector in one specified analytic route;
3. decide whether that vector is a boundary in the chosen field theory.

The result has two theory labels which must not be suppressed:

| Theory                                                | One-loop local Euclidean disposition |
|-------------------------------------------------------|--------------------------------------|
| strict pure Weyl gravity, fixed field content         | obstructed                           |
| formal $\tau$ -adic Wess–Zumino compensator extension | QME restored                         |

The second row does not reverse the first. It changes the BV complex.

All local results carry LOCAL-ALGEBRAIC; the coefficient computation also carries EUCLIDEAN-SPECTRAL. No result in this paper carries LORENTZIAN-CAUSAL.

## 2 Strict local BV obstruction space

The local jet algebra tracks ghost number, antifield number, form degree, engineering dimension, Grassmann parity, spacetime parity, and Weyl weight. Tensor symmetries, algebraic and differential Bianchi identities, graded commutativity, dummy-index relabelling, integration by parts, and total derivatives are reduced before exact sparse linear algebra is performed.

On the regular Bach-locus chart, the minimal Koszul–Tate sector contracts at positive antifield number. Ten nonminimal pairs contract explicitly, and an arbitrary invertible local BV-canonical gauge-fixing transformation preserves the quotient. The pure-diffeomorphism and mixed Diff–Weyl total complexes add no independent four-dimensional anomaly class.

**Theorem 2.1** (Strict dimension-four anomaly quotient on the regular Bach locus). *In the declared complete gauge-fixed local BV complex on a regular Bach-locus chart,*

$$\dim H_{\text{even}}^{1,4}(s \mid d) = 2, \quad \dim H_{\text{odd}}^{1,4}(s \mid d) = 1.$$

*Representatives may be chosen as*

$$[\omega C^2], \quad [\omega E_4], \quad [\omega C\tilde{C}].$$

*The type-D carrier  $\omega \square R$  is exact with its stored  $R^2$  counterterm and horizontal current.*

The qualifier is load-bearing. In adapted jet coordinates the Bach Euler row and its differential consequences replace the transverse-tracefree highest metric jets. Rank-changing strata with additional reducibility or Killing data are not covered by this chart theorem.

The proof is an antifield-number spectral sequence rather than a list of familiar densities, following the local antifield-BRST architecture of Barnich, Brandt, and Henneaux [3]. Its exact steps are:

1. At  $E_0$ , the Koszul–Tate differential contracts the six adapted pairs  $(g^*, E_g)$ ,  $(\xi^*, N_\xi)$ ,  $(\omega^*, N_\omega)$  and their Lie-prolonged partners. Hence every positive-antifield column vanishes on the regular chart.
2. At  $E_1$ , the longitudinal differential reduces the antifield-zero sector to  $\text{Diff} \times \text{Weyl}$  invariant jets. The forward tensor-graph enumerator and reverse signature coverage agree under the dimension-four bound.
3. Relative d-cohomology and the total descent complex leave two even Weyl-ghost classes and one odd class. The pure-Diff and mixed total-complex blocks have zero quotient.
4. Ten pointwise nonminimal pairs contract, and the contraction transports through any invertible local BV-canonical gauge-fixing map. Therefore the minimal, nonminimal, and gauge-fixed quotients are chain-isomorphic.

The spectral sequence collapses at  $E_2$ . In the type-D row the stored primitive and horizontal current obey the local identity

$$\sqrt{g} \omega \square R d^4 x = -\frac{1}{12} s(\sqrt{g} R^2 d^4 x) - d \star (R d\omega - \omega dR).$$

Its integral on a boundaryless carrier is therefore  $\int \sqrt{g} \omega \square R = -(1/12)s \int \sqrt{g} R^2$ . The full ranks and pair counts appear in the generated spectral-sequence table of the computational supplement.

Closure, exact primitives, complete basis manifests, and normalized dual nonmembership witnesses are stored independently. The universal diffeomorphism descent and intrinsic Weyl descent are recorded separately; in particular, a nonzero Diff-horizontal tower is not called a nontrivial intrinsic Weyl descent.

This separation matches the algebraic Weyl-anomaly classification of Boulanger [5]: the Euler/type-A class is distinguished by a nontrivial intrinsic Weyl descent, whereas strictly Weyl-invariant type-B densities have trivial intrinsic Weyl descent. The present calculation has a different target and scope. It retains the universal Diff completion, the dynamical metric antifields, the Koszul–Tate filtration, and the explicit minimal-to-gauge-fixed BV comparison needed for the pure-gravity master equation.

### 3 Coefficient-bearing one-loop breaking

The coefficient route combines two independent curvature carriers. A local Ricci-flat carrier exposes  $C^2$ , while round  $S^4$  fixes the Euler coordinate. The first five-dimensional bundle in the Euclidean symbol sequence is the gauge-parameter bundle  $T \oplus \mathbf{1}$  (four Diff directions plus one Weyl direction), the ten-dimensional middle bundle is  $S^2 T^*$ , and the final five-dimensional bundle is the transverse-tracefree symbol slice. Thus

$$(T \oplus \mathbf{1})_5 \xrightarrow{\sigma(K)} (S^2 T^*)_{10} \xrightarrow{\sigma(P_{\text{TT}})} (S_{\text{TT}}^2 T^*)_5, \quad \text{im } \sigma(K) = \ker \sigma(P_{\text{TT}}).$$

The separate  $5 \leftarrow 10 \leftarrow 5$  sequence is its cotangent formal adjoint; the endpoint is not being identified with the unreduced ten-component Bach equation. Determinant multiplicities, the functional measure, conformal-Killing zero modes, the factorwise spectral cut, the locally constant negative-scalar phase, and the parity Ward identity are included in the accepted integration slice.

The complete local determinant ledger is small enough to display:

| sector                 | operator                | order | rank | statistics | $Z$ exponent | zero modes | local prescription                                 |
|------------------------|-------------------------|-------|------|------------|--------------|------------|----------------------------------------------------|
| metric TT, upper       | $\Delta_{2,\perp}(4)$   | 2     | 5    | bosonic    | $-1/2$       | 0          | factorwise cut                                     |
| Diff–Weyl scalar ghost | $\Delta_0(-4)'$         | 2     | 1    | fermionic  | $+1/2$       | 5          | cut;<br>negative-mode<br>phase locally<br>constant |
| metric TT, lower       | $\Delta_{2,\perp}(2)$   | 2     | 5    | bosonic    | $-1/2$       | 0          | factorwise cut                                     |
| transverse Diff ghost  | $\Delta_{1,\perp}(-3)'$ | 2     | 3    | fermionic  | $+1/2$       | 10         | factorwise cut                                     |

All four factors use the same factorwise zeta/heat-kernel spectral cut. The single negative scalar phase is locally constant on the fixed stabilizer stratum and therefore drops out of the local  $b_4$  variation; its global phase is not claimed. The primes remove precisely the fifteen conformal-Killing ghost modes. York/Hodge Jacobians and the ten nonminimal doublets have already been included before the four displayed factors are formed.

The same ledger carries the following exact local heat-kernel coordinates after determinant exponents, measure factors, and quartet cancellations:

| sector                 | $C^2$  | $E_4$    | $C\tilde{C}$ | $\square R$ |
|------------------------|--------|----------|--------------|-------------|
| metric TT, upper       | 37/24  | -31/72   | 0            | 0           |
| Diff-Weyl scalar ghost | 89/120 | -269/360 | 0            | 0           |
| metric TT, lower       | 29/8   | -181/72  | 0            | 0           |
| transverse Diff ghost  | 29/40  | -79/120  | 0            | 0           |
| exact sum              | 199/30 | -87/20   | 0            | 0           |

The first two columns are reconstructed factor by factor from two independent carriers: the Ricci-flat coefficient gives  $c_i - a_i$ , while the round- $S^4$  calculation gives  $a_i$ . Thus  $c_i = (c_i - a_i) + a_i$  for every row, not merely after summation. The parity-odd zero is factorwise because every operator is a real parity-even tensor Laplacian. By contrast, the displayed  $\square R$  zero is a convention for the *total* local functional: the single allowed  $R^2$  counterterm sets the summed type-D coordinate to zero. It is not four independently scheme-invariant factorwise  $\square R$  calculations. This is still a bosonic-Weyl-parameter Ward carrier, not yet a BV cohomology class. The next step performs the regulated Ward insertion and ghost replacement.

To make the determinant-to-BV bridge explicit, let  $\sigma$  denote a bosonic Weyl parameter. Dimensional regularization first gives the local supertrace

$$\Gamma_{\text{div}}^{(1)}(d) = \frac{\mu^{d-4}}{(4\pi)^2(d-4)} \int \sqrt{g} [cC^2 - aE_4 + b\square R + \text{BRST-exact rows}].$$

The regulated Ward insertion is not inferred from this scalar functional by name matching. It is computed as

$$\mathcal{S}_S \Gamma_{\text{reg}}^{(1)} = \lim_{d \rightarrow 4} s_d \Gamma_{\text{div}}^{(1)}(d),$$

where the  $(d-4)$  Weyl variation of the dimensionally continued densities cancels the pole. Replacing  $\sigma$  by the Weyl ghost  $\omega$ , adding the certified universal Diff completion, and reducing the exact, gauge-dependent, and total-derivative rows against the complete gauge-fixed  $H^{1,4}(s \mid d)$  gives

$$\mathcal{S}_S \Gamma_{\text{reg}}^{(1)} = \frac{1}{(4\pi)^2} \int \sqrt{g} \omega (cC^2 - aE_4 + b\square R) + s\mathcal{B} + d\mathcal{C}.$$

More explicitly, for the continued integrated carriers  $I_i(d)$  used by the regulator, the cohomological residue identity has the form

$$s_d I_i(d) = (d-4)A_i^4 + s_d B_i(d) + dC_i(d) + O((d-4)^2),$$

where  $A_i^4$  is the Diff-completed ghost insertion represented at top form by  $\omega C^2$ ,  $\omega E_4$ , or  $\omega \square R$ . Multiplication by the pole and passage to the quotient therefore sends each heat-kernel coordinate to the corresponding normalized dual coordinate in  $H^{1,4}(s \mid d)$ . This residue map, rather than equality of familiar coefficient names, is the determinant-to-Slavnov bridge. This is the coefficient-bearing map from the Euclidean integration slice to a ghost-number-one Slavnov class; it is why a logarithmic divergence, trace anomaly, and BV anomaly are not silently conflated. The complete reduction can be summarized by the sequence

$$\begin{array}{ccc} \Gamma_{\text{div}}^{(1)}(d) & \xrightarrow{s_d \text{ and } \text{Res}_{d=4}} & \mathcal{W}_\sigma^{(1)} \\ & & \downarrow \sigma \mapsto \omega + \text{Diff completion} \\ & & Z^{1,4}(s \mid d) \longrightarrow H^{1,4}(s \mid d). \end{array}$$

Here  $\mathcal{W}_\sigma^{(1)}$  is the bosonic Ward insertion, the lower-left object is a ghost-number-one BV cocycle, and only its image under the final quotient arrow is the anomaly class. That last arrow removes explicit Slavnov-exact and horizontal rows. No determinant factor or heat-kernel coordinate is itself promoted directly to the class.

**Theorem 3.1** (Repository one-loop Slavnov breaking). *In the normalization*

$$S_W = \alpha_C \int \sqrt{g} C^2,$$

*the repository-regulated breaking has coordinates*

$$(C^2, E_4, C\tilde{C}, \square R) = \left( \frac{199}{30}, -\frac{87}{20}, 0, 0 \right).$$

*After reduction against the complete strict quotient,*

$$\mathcal{A}^{(1)} = \frac{1}{(4\pi)^2} \left( \frac{199}{30} [\omega C^2] - \frac{87}{20} [\omega E_4] \right).$$

*Consequently the strict fixed-field-content local Euclidean QME is obstructed at one loop.*

The odd zero is derived from the parity Ward identity for the declared real tensor-Laplacian regulator. The type-D zero is scheme-dependent but exact; it cannot cancel either nontrivial coordinate. A beta function alone would not prove this theorem: the decisive object is the regulated Slavnov breaking reduced in the complete BV quotient.

The even numbers agree, after translating the Euler-sign convention, with the classic one-loop Weyl-gravity anomaly of Fradkin and Tseytlin [6]. Our claim is not that these constants were previously unknown. It is the content-addressed identification of the repository elliptic complex, multiplicities, measure, zero modes, contour, and regulator with a complete BV obstruction basis, followed by the explicit Slavnov-class disposition. The local  $\square R$  term has a separate scheme-sensitive history; the recent heat-kernel analysis of Barvinsky et al. [8] makes that distinction explicit. In the scheme frozen here its coefficient is zero and the class is in any case removed by the stored  $R^2$  primitive.

**Proposition 3.2** (Standard unitary free matter cannot cancel the vector). *Let the added matter content be any nonnegative combination of real conformal scalars, Weyl or Dirac fermions, and standard gauge vectors. No such combination cancels both even coordinates of the strict breaking.*

These are free, massless spectator fields conformally coupled to the same metric; gauge vectors include their standard BRST ghosts. In the convention  $(c, -a, p, b_{\square R})$  their exact vectors are

| field                 | $c$    | $-a$    | $p$ | $b_{\square R}$ |
|-----------------------|--------|---------|-----|-----------------|
| pure Weyl graviton    | 199/30 | -87/20  | 0   | 0               |
| real conformal scalar | 1/120  | -1/360  | 0   | 1/180           |
| Weyl fermion          | 1/40   | -11/720 | 0   | 1/60            |
| Dirac fermion         | 1/20   | -11/360 | 0   | 1/30            |
| gauge vector          | 1/10   | -31/180 | 0   | -1/10           |

The type-D column is the common reference heat-kernel scheme and is not used to cancel either nontrivial coordinate. The Weyl row uses the common diffeomorphism-, Lorentz-, and gauge-Ward-preserving determinant, for which the odd coordinate vanishes and the even anomaly is one half the

Dirac row. The coefficients agree between an exact minimal-operator heat-kernel ledger, including the vector ghost complex, and an independent index/central-charge reconstruction. A Yang–Mills adjoint contributes  $\dim(\text{ad } G)$  complete vector rows. An ordinary homogeneous conformal compensator with the scalar kinetic operator is exactly the scalar row. A shifting Wess–Zumino compensator is not such a row: it changes the BV complex and counterterm algebra.

The separating functional is simply  $\ell(c, -a) = c$ : it is strictly positive on the gravity vector and on every allowed matter ray. Hence no nonnegative real combination, and a fortiori no nonnegative integer field content, can reach the origin. Interacting fixed points, wrong-sign or nonunitary matter, conformal higher spins, and compensator extensions lie outside this no-go.

The complete formal signed determinant lattice is nevertheless explicit. If  $N_V = N_{U(1)} + \sum_i \dim(\text{ad } G_i) N_{\text{YM},i}$ , every signed solution is

$$N_s = 128 + 48t, \quad N_W = -308 - 2u - 20t, \quad N_D = u, \quad N_V = t,$$

with  $u, t \in \mathbb{Z}$ . The Smith invariant factors are  $(1, 30)$  and the homogeneous kernel is generated by  $(0, -2, 1, 0)$  and  $(48, -20, 0, 1)$ . Negative multiplicity means an inverse determinant or wrong-statistics power, not healthy standard-sign matter. A shifting Wess–Zumino compensator is not a lattice column: it changes the BV complex and local counterterm algebra.

**Corollary 3.3** (Gauge representations cannot rescue the healthy cone). *Assign any finite healthy field content above to finite-dimensional representations of compact gauge groups and impose arbitrary local cubic, mixed gauge–gravitational and global anomaly constraints. The simultaneous strict Weyl- and gauge-anomaly solution set is empty.*

Indeed, forgetting the gauge labels multiplies each field row by the positive integer dimension of its representation and maps every such assignment into the nonnegative matter cone just proved empty. Gauge-anomaly conditions only shrink the domain. Vectorlike pairs can cancel chiral gauge anomalies but add two positive  $C^2$  contributions. No bounded group or highest-weight scan is needed, and none is claimed: this corollary does not select a gauge group, Standard Model, GUT, particle spectrum or ultraviolet completion.

**Proposition 3.4** (Paneitz changed-field-content ray). *One real four-dimensional Paneitz scalar has the exact anomaly vector*

$$(c, -a, p, b_{\square R}) = \left( -\frac{1}{15}, \frac{7}{90}, 0, \frac{1}{15} \right)$$

*in the declared raw higher-order heat-kernel scheme. After quotienting the exact type-D coordinate, the enlarged nonnegative multiplicity cone contains the solution*

$$N_V = 61, \quad N_{P_4} = 191,$$

*with every other standard-matter multiplicity zero. This is not healthy standard matter: the Paneitz propagator has opposite second-order residues.*

The four coordinates agree between two exact routes. The first is the local dimensionally continued Juhl–Gilkey heat coefficient completed by the Branson–Orsted conformal principle; the second combines the Einstein factorization  $P_4 = \Delta(\Delta + R/6)$  with the multiplicative-anomaly-improved Casimir relation of [12, 13]. Both give

$$a = -\frac{7}{90}, \quad c - a = \frac{1}{90}, \quad \gamma_{\Delta J} = -\frac{32}{45}, \quad b_{\square R} = \frac{4a - \gamma}{6} = \frac{1}{15}.$$

The type-D value is displayed for scheme reconciliation but remains an exact class.

The projected integer lattice has Smith invariant factors  $(1, 30)$  and

$$N_s = 128 - 16p + 48v, \quad N_W = -308 - 2d - 20v + 8p, \quad N_D = d, \quad N_V = v, \quad N_{P_4} = p.$$

The inequalities  $p \leq 8 + 3v$  and  $p \geq \lceil (308 + 20v)/8 \rceil$  prove that  $v = 61, p = 191$  is the first solution by vector count. In the common raw  $\square R$  scheme the three-coordinate nonnegative cone is still empty, with exact separator  $\ell = -3c - 5(-a) + 3b_{\square R}$ . This raw statement is scheme bookkeeping, not another nontrivial anomaly.

No conformal higher-spin gauge column is inferred from this scalar result. Spin two is the strict Weyl graviton already under study, and the next new gauge candidate has no declared complete off-shell BV/reducibility complex, nonminimal gauge fixing, generic-background elliptic operator and two-route raw coefficient payload in the repository. The higher-derivative family therefore remains open beyond the Paneitz scalar.

**Proposition 3.5** (First new conformal gauge-carrier obstruction). *Fix the finite carrier contract consisting of an off-shell field-ghost-antifield and reducibility dictionary, nilpotent minimal differential, exact Noether identity, nonminimal gauge fixing, generic-background elliptic operator and domain, chirality and zero-mode ledger, determinant powers and contours, two independent raw coefficient routes, and a kinetic-sign audit. Neither the minimal-depth conformal spin-3/2 field nor the minimal-depth bosonic conformal spin-3 field satisfies this contract with the presently available local data. Hence neither defines an anomaly-lattice column.*

For spin 3/2, the explicit Weyl-corrected quadratic action has gauge variation proportional to the Bach tensor [16, 14]. Its Noether identity therefore closes on Bach-flat backgrounds but not on the declared generic background. For minimal-depth spin 3, the pure sixth-order block fails beyond first order in background curvature: the curvature-shifted spin-one sector participates explicitly in the gauge identity [17]. A separate superconformal-completion argument indicates an additional shifted spin-two sector [16]; we do not promote that argument to an independent nonsupersymmetric necessity theorem. No closed all-curvature minimal-depth spin-1-spin-3 BV/elliptic carrier, nor a disposition of the indicated spin-two sector, is imported. The known scalar-ghost maximal-depth spin-three model of [15] is a different gauge system and is not substituted for it.

This is a fail-closed carrier obstruction, not a higher-spin no-go theorem. In particular, an absent carrier is not a zero anomaly coefficient. The strict receiver and mutation tests are recorded in `FIRST_NEW_CONFORMAL_GAUGE_FIELD_CARRIER_OBSTRUCTION.json`.

## 4 The compensator cotangent lift

Adjoin a scalar with BRST transformation

$$Q\tau = \mathcal{L}_\xi \tau + \omega.$$

Its cotangent rows are not guessed. They are the Euler derivatives of the single minimal master term

$$S_{\tau, \min} = \int \tau^* (\mathcal{L}_\xi \tau + \omega).$$

In addition to the field row this forces

$$\delta\omega^* = 2g_{\mu\nu}g^{*\mu\nu} + \tau^*, \quad \delta\xi_\mu^* = N_{\xi, \mu} + \tau^* \nabla_\mu \tau, \quad \gamma\tau^* = \mathcal{L}_\xi \tau^*,$$

together with the opposite-adjoint Lie-prolonged rows.



**Theorem 4.1** (Exact minimal-BV cotangent lift). *On every strict-plus-compensator generator and covariant jet generator,*

$$\delta^2 = 0, \quad \delta\gamma + \gamma\delta = 0, \quad Q^2 = 0.$$

*The canonical change*

$$\hat{g} = e^{-2\tau}g, \quad \hat{g}^* = e^{2\tau}g^*, \quad \hat{\tau}^* = \tau^* + 2g_{\mu\nu}g^{*\mu\nu}$$

*preserves the canonical one-form and sends the Weyl Koszul–Tate row to  $\delta\omega^* = \hat{\tau}^*$ . Thus  $(\tau, \omega, \omega^*, \hat{\tau}^*)$  is an exact contractible quartet.*

In the ordered basis  $(\tau, \omega, \omega^*, \hat{\tau}^*)$ , the Weyl part of the differential and its homotopy act by

$$Q_W\tau = \omega, \quad Q_W\omega^* = \hat{\tau}^*, \quad h\omega = \tau, \quad h\hat{\tau}^* = \omega^*,$$

and vanish on the other two basis generators. Consequently

$$Q_W h + h Q_W = N_{\text{quartet}}.$$

On a monomial of positive quartet number the contracting homotopy is  $N_{\text{quartet}}^{-1}h$ , extended to covariant jets as a graded derivation. The Diff and horizontal pieces preserve quartet number, so the filtered total-complex contraction follows by homological perturbation in the declared formal completion.

For clarity, the total-complex contraction can be read without replaying the boundary matrix. Write  $\mathcal{D} = Q_W + \delta$ , with  $\delta$  the remaining Diff, Koszul–Tate, and horizontal rows in dressed variables. The four quartet members occur in matched covariant Diff modules, and  $h$  is extended to their jets as a covariant graded derivation. Hence  $[N_{\text{quartet}}, \delta] = 0$  and the graded anticommutator  $[\delta, h]_+$  vanishes. Therefore

$$\mathcal{D}h + h\mathcal{D} = N_{\text{quartet}}.$$

If  $\mathcal{D}\mathbf{a} = 0$  and  $\mathbf{a}_n$  has positive quartet number  $n$ , then  $\mathbf{a}_n = \mathcal{D}(n^{-1}h\mathbf{a}_n)$ . The total cohomology is consequently represented at quartet number zero, where  $\mathcal{D}$  is the pure-Diff BV total differential of  $\hat{g}$ . Thus  $H_{\text{ext}}^{1,4} = 0$  imports the independently certified pure-Diff ghost-one result; it is not inferred only from the four displayed Wess–Zumino primitives.

The exponential change is an automorphism only after declaring the formal  $\tau$ -adic local analytic completion. Its two inverse series are replayed with exact rational coefficients and obey at every order

$$\sum_{k=0}^n \frac{(-2)^k}{k!} \frac{2^{n-k}}{(n-k)!} = \frac{2^n}{n!} (1-1)^n = \delta_{n0}.$$

No finite-polynomial-in- $\tau$  theorem is inferred.

## 5 Extended cohomology and QME restoration

Quartet contraction reduces the extended complex to pure Diff BV cohomology in the dressed metric  $\hat{g}$ . The dimension-four counterterm inventory is therefore larger than in strict Weyl gravity:  $R(\hat{g})^2$  is now closed. The declared algebra is the formal  $\tau$ -adic local analytic jet completion generated by  $\hat{g}$ , its cotangent variables, Diff ghosts, the quartet, curvatures and covariant derivatives, with the same dimension and form-degree bounds as the strict calculation. The canonical dressed change is invertible in this completion. After quartet contraction, forward invariant generation and reverse grading-signature coverage leave precisely the pure-Diff dimension-four metric invariants below; the independently completed pure-Diff ghost-one total complex has zero quotient. Nonminimal and gauge-fixing transport then preserve that result. Thus the vanishing assertion is a complete statement in this declared formal algebra, not an AFN0 primitive test alone.

**Theorem 5.1** (Formal  $\tau$ -adic extended quotient). *In the declared formal  $\tau$ -adic local analytic jet algebra on the regular Bach locus,*

$$H_{\text{ext,even}}^{0,4} = \text{span}\{C(\widehat{g})^2, E_4(\widehat{g}), R(\widehat{g})^2\},$$

$$H_{\text{ext,odd}}^{0,4} = \text{span}\{C(\widehat{g})\widetilde{C}(\widehat{g})\}, \quad H_{\text{ext}}^{1,4} = 0.$$

$\square R(\widehat{g})$  remains horizontally exact. There is no incoming dimension-four positive-antifield boundary: the only dimension-zero natural symmetric tensor is  $\widehat{g}$ , whose image is the vanishing trace of the Bach row.

Explicit primitives for the strict four-candidate anomaly carrier are

$$B_C = \int \sqrt{g} \tau C^2, \quad B_P = \int \sqrt{g} \tau C \widetilde{C},$$

and the Euler Wess–Zumino functional is

$$B_E = \int \sqrt{g} [\tau E_4 + 4G^{\mu\nu} \nabla_\mu \tau \nabla_\nu \tau - 4(\square \tau)(\nabla \tau)^2 + 2(\nabla \tau)^4].$$

With the conventions above,

$$s_{\text{ext}} B_E = \int \sqrt{g} \omega E_4 \pmod{\mathfrak{d}}.$$

The termwise intrinsic Euler descent and its universal Diff completion are stored separately. Together with the  $R^2$  primitive  $B_\square$ , these four primitives have an exact rank-four boundary matrix. The curved-background Euler primitive agrees with the standard dilaton Wess–Zumino construction obtained from finite Weyl transformation of dimensionally regulated counterterms [7]. The new certificate needed here is not the displayed AFN0 formula alone: it is the cotangent lift, quartet contraction, exhaustive extended  $H^{0,4}/H^{1,4}$  comparison, and coefficient-bearing BV boundary equation.

This restoration has a physical price. The compensator theory is not quantum-equivalent to strict Weyl gravity: gauge fixing  $\tau = 0$  does not turn the anomalous strict quotient into an anomaly-free theory, while the dressed Diff theory permits counterterms such as  $R(\widehat{g})^2$  and finite normalizations unavailable in the strict quotient. Exactness is obtained by enlarging both the BV complex and its local counterterm algebra.

**Corollary 5.2** (One-loop local Euclidean QME restoration). *The counterterm*

$$-\frac{\hbar}{(4\pi)^2} \left( \frac{199}{30} B_C - \frac{87}{20} B_E \right)$$

satisfies

$$\mathcal{A}^{(1)} + s_{\text{ext}} S_{\text{ct}}^{(1)} = 0 \pmod{\mathfrak{d}}.$$

Hence the compensator-extended local Euclidean QME is restored at one loop in the declared formal  $\tau$ -adic algebra.

**Theorem 5.3** (Conditional all-order formal local restoration). *Let the coefficient ring be completed in*

$$\mathfrak{m} = (\hbar, z_C, z_E, z_{R^2}, z_P),$$

and retain homogeneous engineering-dimension-four local functionals in the formal  $\tau$ -adic analytic jet algebra. Assume an all-order quantum action principle with the following properties: after

restoration through order  $n - 1$ , the first order- $n$  breaking is a local ghost-one element of this algebra; it obeys the linearized Wess–Zumino consistency condition modulo  $d$ ; subtraction is continuous in the  $\mathfrak{m}$ -adic filtration; and the regular Koszul–Tate chart persists under the formal coupling deformation.

Under these hypotheses the compensator-extended local QME can be restored at every finite formal order. Its complete stable counterterm module is

$$\begin{aligned} H_{\text{ext,even}}^{0,4} &= \text{span}\{C(\widehat{g})^2, E_4(\widehat{g}), R(\widehat{g})^2\}, \\ H_{\text{ext,odd}}^{0,4} &= \text{span}\{C(\widehat{g})\widetilde{C}(\widehat{g})\}, \end{aligned}$$

modulo  $d$ -exact terms, while exact ghost-zero terms generate local canonical transformations. In particular,  $R(\widehat{g})^2$  is an indispensable independent coupling direction.

*Proof.* The imported extended calculation is antifield-complete and gives  $H_{\text{ext}}^{1,4} = 0$  in both parities. A counterterm deformation  $\delta_{\text{ct}}$  raises the complete filtration, so the regular Koszul–Tate contraction persists by the formal homological-perturbation inverse

$$(1 + h\delta_{\text{ct}})^{-1} = \sum_{k \geq 0} (-h\delta_{\text{ct}})^k.$$

Thus the zero ghost-one  $E_1$  page remains zero after every formal coupling deformation. If  $\mathcal{A}_n$  is the first order- $n$  breaking, the quantum action principle and consistency give

$$s_{\text{ext},n-1}\mathcal{A}_n + d a_n = 0.$$

Vanishing relative cohomology yields

$$\mathcal{A}_n = s_{\text{ext},n-1}B_n + dC_n.$$

Adding  $-B_n$  removes the breaking. The closed ambiguity lies in the displayed  $H^{0,4}$  module and is absorbed by coupling redefinitions, total derivatives, and canonical transformations. Hence the induction repeats at every finite formal order. The exact recurrence and its independent replay are stored in `TAU_ADIC_ALL_LOOP_LOCAL_QME_STABILITY`.  $\square$

This is the standard algebraic-renormalization induction based on local BRST cohomology [3]. The theorem is conditional: the repository has not constructed an all-order regulator or subtraction scheme that proves the stated quantum action principle on the completed  $\tau$ -adic domain. The infinite  $\tau$  expansions of dressed invariants are formal completions of four independent cohomology generators, not an infinite tower of independent couplings.

## 5.1 Regulator and measure status

The preceding induction remains valid as a cohomological theorem under its declared quantum-action-principle hypothesis. Four exact preflights now explain why that hypothesis has not yet been implemented by an action-independent regulator.

First, ordinary dimensional regularization with minimal subtraction does not close on the strictly four-dimensional stable module. Two Euler continuations may differ by

$$E_d^{(X)} = E_d^{(0)} + \epsilon X + O(\epsilon^2).$$

Because the Euler pole residue is  $-87/20$ , projecting to four dimensions before subtraction erases the nonzero finite shift  $-(87/20)X$ . The common even antifield-zero  $d$ -dimensional geometric premodule is nevertheless exact: it is generated by

$$(R_{\mu\nu\rho\sigma}^2, R_{\mu\nu}^2, R^2, \square R),$$

with a certified rational change of basis to  $(C^2, E_4, R^2, \square R)$ . In particular,

$$\partial_\epsilon C_{4-\epsilon}^2|_{\epsilon=0} = -\frac{1}{2}C^2 + \frac{1}{2}E_4 - \frac{1}{18}R^2.$$

Completion to the full BV module requires the selected action's  $d$ -dimensional Koszul–Tate rows, while the odd Pontryagin direction requires an explicit noninteger-dimensional epsilon-tensor or  $\gamma_5$  prescription.

Second, the dressed canonical transformation is not measure preserving on a finite common carrier. For the ten metric components and their parity-reversed cotangent partners,

$$\text{Ber}_{\text{BV}}^{(N)} = \exp\left(-40 \sum_{i=1}^N \tau_i\right).$$

Its inverse, composition and canonical-one-form identities are exact, but a continuum expression  $-40 \text{Tr}_R(\tau)$  depends on the selected regulator, boundary domain, zero-mode projector and contours. There is no unique action-independent counterterm module absorbing it.

Third, a strictly four-dimensional covariant receiver exists conditionally. For a selected fourth-order elliptic Hessian  $K$  satisfying  $K^\sharp = K$ ,  $[Q, K] = 0$ , and matched boundary and zero-mode projector conditions, proper-time/heat-kernel or higher-derivative plus BV Pauli–Villars routes give

$$\log \text{Ber}_R = -40 \text{Tr}[\Pi_\perp \tau F(K/\Lambda^4) \Pi_\perp].$$

If  $K$  fails to intertwine BRST, the Ward identity retains the corresponding Duhamel insertion containing  $QK$ . This is a receiver theorem, not an instantiated regulator: the selected gauge-fixed Hessian symbol complex, domains, zero modes and real contours remain missing.

The four-dimensional and DR/MS subtraction ambiguities are not identified. The former is parameterized directly in the four-dimensional  $H^{0,4}$  module; the latter first carries an evanescent-continuation torsor. Both need an action-specific mixing map. Candidate A's auxiliary-scalar parent is a scoped classical obstruction after failing its physical-sign and raw- $D$  gates on the frozen carrier. Candidate B's reducible three-form parent is also a scoped classical obstruction: it fails the declared trace/background, global-flux and raw- $D$  gates. Thus neither declared minimal repair selects an action, and neither Hessian is imported into the conditional quantum theorem here.

The subsequent exact family classification strengthens this two-candidate disposition without making it universal. The seven-gate good locus is empty for the complete declared minimal formal polar family with constant real couplings, four metric derivatives, at most two compensator derivatives, one invertible algebraic auxiliary presentation of  $R^2$ , and an optional minimal Henneaux–Teitelboim sector under the small reducible three-form gauge group. The classification does not cover higher-than-two-derivative phase EFT operators, nonconstant phase-dependent couplings, kinetic or nonlinear potentials for the topological multiplier, fixed-flux or fixed-multiplier superselection sectors, large/global three-form quotients, active-clock retunings, or an independent conformal gauge connection; the parity-odd Pontryagin direction is also outside the classified even family. Hence it is a scoped minimal-family no-go, not a universal compensator no-go. No Candidate C action or action hash is selected, so every regulator, Jacobian, mixing and QAP implementation gate remains action dependent and no candidate Hessian enters the conditional quantum theorem.

The independently frozen successor closes one previously excluded direction: the complete shift-symmetric quadratic  $P(X)$  active-clock extension, with

$$X = \hat{g}^{ab} \partial_a \theta \partial_b \theta, \quad P(X) = p_0 + p_1 X + p_2 X^2,$$

and no Henneaux–Teitelboim sector or new fields. On the common unit-cylinder and frozen Berger fixtures its stationary equations have the one-dimensional common stationary locus

$$(\alpha_B, \alpha_R, M_P^2, p_0, p_1, p_2) = t \left( \frac{81}{20}, \frac{27}{3290}, -\frac{324}{1645}, \frac{486}{1645}, \frac{18}{25}, 1 \right).$$

The seven-gate good locus is nevertheless empty for two independently replayed reasons. For every  $t \neq 0$ , a rational congruence puts the coupled gravity–auxiliary velocity form into

$$\text{diag}(-6, 6, -36t/25),$$

so its gravity–auxiliary pair remains split. Independently, the standard-sign cylinder clock requires  $t < 0$  whereas the Berger clock requires  $t > 0$ ; at  $t = 0$  the action vanishes and supplies neither pairing nor dynamics. This is a scoped no-go for the declared quadratic active-clock family, not a universal  $k$ -essence or compensator no-go. It excludes neither higher  $P(X)$  nor higher phase derivatives, nearby backgrounds, fixed-charge sectors, new fields, or enlarged gauge groups. No Candidate  $C_{\text{active}}$  action hash is exported, so regulator and QAP consumers remain action dependent.

The exact background-stability theorem strengthens this pointwise result. Write  $\kappa = r_{\text{cyl}}^{-2}$ , let  $q = c^2$  be the Berger squashing at horizontal scale one, and set  $D\theta = \nu > 0$ . On the rational open box

$$\mathcal{N}_{\square} = \left( \frac{15}{16}, \frac{17}{16} \right)_{\kappa} \times \left( \frac{1}{5}, \frac{1}{4} \right)_q \times \left( \frac{2}{3}, \frac{5}{6} \right)_{\nu},$$

the stacked stationary matrix has rank five. At every point its nonzero solutions form one action-space ray

$$(\alpha_B, \alpha_R, M_P^2, p_0, p_1, p_2) = \lambda K(\kappa, q, \nu), \quad \lambda \neq 0.$$

The couplings therefore vary with the background. This is not an implicit-function theorem for one fixed action.

Two failures are structurally stable throughout  $\mathcal{N}_{\square}$ . The gravity–auxiliary velocity pair retains the exact eigenvalues  $(+3, -3)$ , and the raw- $D$  Hamiltonian retains explicit field-space witnesses with values  $+3$  and  $-3$ . Moreover, for  $\lambda < 0$  the cylinder clock has standard sign while the Berger clock has the wrong energy/charge orientation; for  $\lambda > 0$  the Berger clock is standard-sign, hyperbolic, positive-energy, charged and subluminal while the cylinder clock has the wrong sign. Hence the seven-gate good locus is empty at every point of the open box.

The first boundary bifurcation is also exact. Along

$$\kappa = 1, \quad \nu = \frac{3}{4}, \quad \lambda = 1,$$

the surface  $q = 1/4$  switches off the algebraic  $R^2$  auxiliary presentation and makes the cylinder clock degenerate without lowering the stationary rank at  $\kappa = 1$ . Below it, at  $q = 9/40$ , the exact  $(p_1, P_X, P_X + 2XP_{XX})$  row is

$$\left( \frac{2961}{800}, -\frac{26649}{12800}, -\frac{174699}{12800} \right),$$

so only the Berger clock is healthy. Above it, at  $q = 21/80$ , the row is

$$\left( -\frac{5949}{3200}, -\frac{124929}{51200}, -\frac{184419}{51200} \right),$$

so both clocks are standard-sign and hyperbolic. Crossing the bifurcation therefore repairs only the clock-sign conflict. The split gravity–auxiliary pair and the raw- $D$  both-sign witnesses persist, so the upper witness is not a viable phase.

This is an open-neighbourhood theorem for parameter-dependent stationary action rays, not a fixed-action or generic-background theorem. It covers neither general  $k$ -essence, higher derivatives, fixed-charge quotients, new fields nor enlarged gauge groups, and supplies no complete causal parent, anomaly/QME, particle or unitarity result. In particular, this classical selection no-go does not weaken the separate formal local Wess–Zumino trivialization theorem.

The later exact synthesis makes the complete declared classical selection frontier explicit. In the notation of its machine certificate, the tested union is exactly

$$\begin{aligned} \mathcal{U}_{\text{tested}} = & \mathcal{T}_{\text{passive}} \cup \mathcal{M}_{\text{polar}}^{(4g,2\theta), \text{small HT}} \cup \mathcal{P}_{\leq 2} \cup (\mathcal{P}_{\leq 2} + \mathcal{G}_{\leq 1}) \\ & \cup \mathcal{L}_3^{\text{literal}}(F_{\leq 1}, +F_X) \cup \mathcal{L}_{3b}^{\text{Horndeski}}(F_{\leq 1}, -2F_X) \cup \mathcal{L}_4^{\text{real Weyl, minimal}}. \end{aligned}$$

Here  $\mathcal{T}_{\text{passive}}$  is the strict action with the formal  $\tau$ -adic extension;  $\mathcal{M}_{\text{polar}}$  is the complete declared parity-even four-metric/two-phase-derivative family with optional normalized small-gauge HT sector;  $\mathcal{P}_{\leq 2}$  is quadratic shift-symmetric  $P(X)$ ;  $\mathcal{G}_{\leq 1}$  adds the first nonexact polynomial braiding  $\beta X \hat{\square} \theta$ ; the two Level-3 components keep the literal  $+F_X$  and convention-correct  $-2F_X$  Hessian-square coefficients distinct; and  $\mathcal{L}_4$  is the lowest-order parity-even theory with one real torsion-free Weyl connection and one formal-polar compensator. The exact result is

$$\mathcal{G}_{\text{good}}(\mathcal{C}) = \emptyset \quad \text{for every separately declared component } \mathcal{C} \subset \mathcal{U}_{\text{tested}}.$$

Thus Candidate A, Candidate B, the former Candidate C frontier, quadratic  $P(X)$ , first braiding, both linear- $F$  curvature conventions, and the minimal real-connection step are terminal components of this union, not live action-selection branches inside it.

This is a union theorem, not a closure theorem. It does not test simultaneous nonzero braiding and Horndeski couplings, higher  $F$  or  $G$ ,  $G_5$ /DHOST, extra fields, fixed-flux or changed superselection quotients, changed backgrounds or fixed-charge reductions, arbitrary hybrids, or general metric-affine/complex gauge geometry. The historical complete direct-sum rank-390 causal promotion is superseded by Candidate A's full mixed Hessian; only its narrower trace Schur complement, reduced scalar Green identities, and phase scalar-wave block survive. Since no selected classical action or Hessian exists, neither a determinant nor an action-specific QAP freeze is activated. The separated real scale plus compact internal  $U(1)$  representation is the next preflight only; it is not a selected action and establishes no stationary locus, causal parent, anomaly cancellation, QME, Hadamard state, particle space, positivity, or unitarity theorem.

The four machine receipts are TAU\_ADIC\_DR\_MS\_QAP\_EVANESCENT\_CLOSURE\_OBSTRUCTION, DRESSED\_CANONICAL\_BEREZINIAN\_LOCALITY\_PREFLIGHT, DRESSED\_EVANESCENT\_GEOMETRIC\_B V\_MODULE\_PREFLIGHT, and DRESSED\_FOUR\_DIMENSIONAL\_COVARIANT\_REGULATOR\_PREFLIGHT. The now-authoritative selection frontier is COMPENSATOR\_MINIMAL\_LADDER\_SYNTHESIS\_AFTER\_LEVEL3B\_V1. It imports the fifteen component artifacts by content hash, including the earlier minimal, Candidate-A/B, quadratic- $P(X)$  and open-neighbourhood results; those remain valid scoped subresults but no longer define live successor branches inside  $\mathcal{U}_{\text{tested}}$ . They do not establish a QAP, an unconditional all-loop QME, or any Lorentzian or state result.

**Proposition 5.4** (The restored QME does not determine a unique  $Q_1$ ). *Define the coefficient-bearing one-loop Wess–Zumino functional by*

$$C_1^{\text{WZ}} = -\frac{1}{(4\pi)^2} \left( \frac{199}{30} B_C - \frac{87}{20} B_E \right).$$

*The local Wess–Zumino Hamiltonian contribution*

$$Q_1^{\text{WZ}} = (C_1^{\text{WZ}}, \cdot)$$

is coefficient-bearing and fixed by the counterterm above. The complete renormalized first Slavnov operator is nevertheless underdetermined. On the dimensionless flat Euclidean fixture  $p = (1, 0, 0, 0)$ , in units of the chosen reference momentum, with  $h_{11} = 1$ ,  $h_{22} = -1$  and all other  $TT$  components zero, together with one conformal polarization, the bulk quadratic response of the allowed basis  $(C(\hat{g})^2, E_4(\hat{g}), R(\hat{g})^2, C(\hat{g})\tilde{C}(\hat{g}))$  is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 \end{pmatrix}.$$

Thus the  $C^2$  and  $R^2$  finite-counterterm directions change the bulk Hamiltonian vector field independently. This response calculation alone does not supply a nonlocal representative, the renormalized BV Laplacian or time-ordered product, or finite normalization conditions. The next proposition supplies the anomaly-induced nonlocal component, but not the Weyl-invariant finite remainder.

## 6 Consequences for one-loop effective-action representatives

**Proposition 6.1** (Anomaly-induced Euclidean representative). *Set*

$$\mathcal{E}_4 = E_4 - \frac{2}{3}\square R, \quad \Delta_4 = \square^2 + 2R^{\mu\nu}\nabla_\mu\nabla_\nu - \frac{2}{3}R\square + \frac{1}{3}(\nabla^\mu R)\nabla_\mu.$$

This is the four-dimensional Paneitz operator [11]. On an invertible Euclidean boundary problem, or on the compatible source sector of a self-adjoint generalized inverse  $G_4$  of  $\Delta_4$ , one anomaly-induced representative is

$$\Gamma_{1,\text{anom}} = \frac{1}{(4\pi)^2} \left\{ \frac{1}{8} \langle \mathcal{E}_4, G_4 (2cC^2 - a\mathcal{E}_4) \rangle + \frac{a}{18} \int \sqrt{g} R^2 \right\}, \quad (c, a) = \left( \frac{199}{30}, \frac{87}{20} \right).$$

Its Weyl variation is the certified anomaly  $(4\pi)^{-2}(cC^2 - aE_4)$  in that sector. The three functional coefficients are

$$\left( \frac{199}{120}, -\frac{87}{160}, \frac{29}{120} \right).$$

The last coefficient follows from  $\delta_\sigma \int \sqrt{g} R^2 = -12 \int \sqrt{g} \sigma \square R$  and is required to return from the canonical  $\mathcal{E}_4$  representative to the repository  $\square R = 0$  convention. This is the Riegert construction [10] with the scheme conversion made explicit; it is not the complete finite effective action.

**Proposition 6.2** (Universal flat-TT logarithmic form factor). *On the positive nonzero-momentum flat Euclidean transverse-traceless carrier, the quadratic conformal-sector form factor is*

$$\Gamma_{1,\text{TT}}^{(2)} = \frac{1}{(4\pi)^2} \langle C^{(1)}, F_C(p^2; \mu) C^{(1)} \rangle, \quad F_C(p^2; \mu) = -\frac{199}{60} \log \frac{p^2}{\mu^2} + z_C(\mu).$$

The logarithmic coefficient obeys

$$-\frac{199}{60} = -\frac{c}{2} = -\frac{\beta_2}{4}, \quad (c, \beta_2) = \left( \frac{199}{30}, \frac{199}{15} \right),$$

and therefore

$$\mu \partial_\mu F_C = \frac{199}{30}, \quad F_C(p^2; \mu) - F_C(q^2; \mu) = -\frac{199}{60} \log \frac{p^2}{q^2}.$$

The factor of 1/2 is fixed by the stated normalization:  $\langle C^{(1)}, C^{(1)} \rangle = 1$  and  $\mu \partial_\mu \log(p^2/\mu^2) = -2$ , so matching the scale response  $c$  requires  $-2A_{\log} = c$  and hence  $A_{\log} = -c/2 = -199/60$ . This fixes

the universal scale-dependent part and the scheme-independent momentum difference. A reference condition  $F_C(\kappa_{\text{ref}}^2; \mu) = N_C$  instead fixes the still-free local constant as

$$z_C(\mu) = N_C + \frac{199}{60} \log \frac{\kappa_{\text{ref}}^2}{\mu^2}.$$

The separation of anomaly-induced and conformal parts and the appearance of logarithmic non-local curvature form factors follow the covariant effective-action framework of Barvinsky and Wachowski [18]. The present certificate is restricted to the declared flat carrier. The next two propositions supply its curvature-squared covariantization and one exact Fradkin–Vilkovisky completion of that selected carrier. They do not compute the third-curvature repository form-factor functions or coefficients, a zero-momentum definition, Lorentzian branch prescription, or a particle state.

**Proposition 6.3** (Endomorphism-class covariant logarithm through curvature order two). *Let  $\Delta_C$  be a positive self-adjoint Laplace-type operator on the algebraic Weyl-tensor bundle, with a fixed self-adjoint elliptic domain, and let  $\Pi_\perp$  project off its kernel. Covariant perturbation theory through total curvature order two gives*

$$\Gamma_{1,C^2 \log}^{(2)} = -\frac{1}{(4\pi)^2} \frac{199}{60} \left\langle C, \Pi_\perp \log \frac{\Delta_C}{\mu^2} \Pi_\perp C \right\rangle.$$

It obeys

$$\mu \partial_\mu \log(\Delta_C/\mu^2) = -2\Pi_\perp,$$

and hence has the certified scale response  $(4\pi)^{-2}(199/30)\langle C, \Pi_\perp C \rangle$ .

This quadratic-curvature functional is independent, at the declared order, within the class of operators sharing the same principal symbol, connection, self-adjoint domain, and source-complement prescription and differing at first curvature order only by a bundle endomorphism. Indeed, if  $\Delta'_C = \Delta_C + V_1 + O(\mathcal{R}^2)$  with  $V_1$  a self-adjoint curvature-order-one bundle endomorphism on the same domain, then

$$d \log(\Delta_C)[V_1] = \int_0^\infty (\Delta_C + s)^{-1} V_1 (\Delta_C + s)^{-1} ds.$$

Sandwiching this variation between two Weyl tensors has curvature order  $1 + 1 + 1 = 3$ . Thus the first missing completion data for this  $C^2$  logarithmic carrier are cubic in the curvature [19, 22]. No independence under a change of bundle connection is claimed; such a change can introduce first-order differential terms not covered by  $V_1$ . Finite  $C^2$  and absolute dressed  $R(\hat{g})^2$  normalizations, third-curvature repository functions and coefficients, and global kernel data remain open. The following proposition supplies a Weyl-orbit completion only for the selected  $C^2$  logarithmic carrier.

**Proposition 6.4** (Fradkin–Vilkovisky completion of the selected logarithmic carrier). *Suppose the Euclidean conformal scalar operator*

$$L_g = \square_g - \frac{1}{6} R_g$$

has a normalized inverse on an asymptotically flat or fixed-boundary function space, with no normalized zero mode. Define

$$u_g = 1 + \frac{1}{6} L_g^{-1} R_g, \quad \Sigma_{\text{FV}}[g] = -\log u_g, \quad \bar{g}[g] = u_g^2 g.$$

Then  $L_g u_g = 0$  and

$$R[\bar{g}] = u_g^{-3} (R_g - 6\square_g) u_g = 0.$$



For boundary-compatible  $g' = e^{2\sigma}g$ , conformal covariance and the normalization imply  $u_{g'} = e^{-\sigma}u_g$ , hence  $\bar{g}[g'] = \bar{g}[g]$ . Therefore

$$\Gamma_{1,C^2}^{\text{conf}}[g] = -\frac{1}{(4\pi)^2} \frac{199}{60} \left\langle C[\bar{g}], \Pi_{\bar{g}} \log \frac{\Delta_C[\bar{g}]}{\mu^2} \Pi_{\bar{g}} C[\bar{g}] \right\rangle_{\bar{g}}$$

is exactly invariant under the declared local Weyl transformations. Every tensor, pairing, projector, operator domain and spectral logarithm is evaluated on the same representative. This is a nonlocal functional invariant under local Weyl transformations, not a local functional.

Since  $\Sigma_{\text{FV}} = -(1/6)\square^{-1}R + O(\mathcal{R}^2)$ , this completion agrees with the preceding carrier through curvature order two and its first forced correction is cubic. The Fréchet derivative of the spectral logarithm identifies that correction carrier, including transported pairing and projector terms, but does not determine the derivative-decorated nonlocal cubic Weyl-invariant form factors of the full one-loop action [18]. If  $L_g$  has a kernel, an orbit-compatible projector and normalization are additional data and the proposition does not apply as stated.

Finally,  $\bar{g}[g]$  is not the compensator-dressed  $\hat{g}[g, \tau] = e^{-2\tau}g$ : the former is a nonlocal functional of the strict metric fixed by scalar-flat gauge, while  $\tau$  is an independent local field in the enlarged formal BV theory.

**Proposition 6.5** (FV anomaly action and dependent Ricci sector). *Write  $\mathcal{E}_4 = E_4 - \frac{2}{3}\square R$  and use the same normalized FV orbit coordinate as above. The exact coefficient-bearing anomaly action is*

$$\Gamma_{1,\text{FV}} = \frac{1}{(4\pi)^2} \int \sqrt{g} \left[ \left( \frac{199}{30} C^2 - \frac{87}{20} \mathcal{E}_4 \right) \Sigma_{\text{FV}} + \frac{87}{10} \Sigma_{\text{FV}} \Delta_4 \Sigma_{\text{FV}} + \frac{29}{120} R^2 \right].$$

Under  $\delta_\sigma \Sigma_{\text{FV}} = \sigma$ , its non-target response cancels coefficientwise:

$$-\frac{87}{5} + \frac{87}{5} = 0, \quad \frac{29}{10} - \frac{29}{10} = 0.$$

Hence

$$\delta_\sigma \Gamma_{1,\text{FV}} = \frac{1}{(4\pi)^2} \int \sqrt{g} \sigma \left( \frac{199}{30} C^2 - \frac{87}{20} E_4 \right).$$

The specialization  $\Sigma_{\text{RFT}} = \frac{1}{4}G_4\mathcal{E}_4$  reproduces exactly  $(199/120, -87/160, 29/120)$  in the earlier Paneitz/Riegert basis.

Moreover, in the declared massless conformal FV decomposition

$$\Gamma_1[g] = \Gamma_{1,\text{FV}}[g] + W_1[\bar{g}[g]], \quad R[\bar{g}] = 0,$$

the conformal functional  $W_1$  is generated by Weyl-curvature invariants before evaluation on  $\bar{g}$ . Therefore terms containing a Ricci scalar in a generic original-metric basis are determined by  $\Gamma_{1,\text{FV}}$  and the re-expansion of  $W_1[\bar{g}]$ . A term displayed as  $RF(\square)R$  may occur in that basis, but its form factor is not a separately specifiable datum in this decomposition [18]. This statement is restricted to the normalized asymptotically-flat or fixed-boundary FV domain and does not extend without new analysis to massive, explicitly nonconformal, or symmetry-broken theories.

**Proposition 6.6** (Algebraic cubic Weyl carriers in four dimensions). *In the pointwise zero-derivative sector of three algebraic Weyl tensors in four Euclidean dimensions, the parity-even and parity-odd carrier spaces are both one-dimensional:*

$$I_e = C_{ab}{}^{cd} C_{cd}{}^{ef} C_{ef}{}^{ab}, \quad I_o = (*C)_{ab}{}^{cd} C_{cd}{}^{ef} C_{ef}{}^{ab}.$$

Equivalently, after the Hodge decomposition on two-forms, the chiral basis is  $I_+ = \text{tr}(C_+^3)$  and  $I_- = \text{tr}(C_-^3)$ , with

$$\begin{pmatrix} I_+ \\ I_- \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} I_e \\ I_o \end{pmatrix}.$$

The exact contraction enumeration has fifteen raw six-slot pairings, three identical-factor orbits, two tracefree-zero orbits, and one surviving triangle  $\text{tr}(C_\pm^3)$  in each chirality. The mixed allocations vanish because the chiral projectors are orthogonal and the resulting lone block can only close through its zero trace. An exact rank-two evaluation witness and the independent Schouten-zero Weyl quotient verify the parity basis.

This is a LOCAL-ALGEBRAIC carrier classification, not a third-order effective-action calculation. Since  $C^3$  has engineering dimension six, it is not an additional dimension-four one-loop local counterterm. The derivative-decorated nonlocal carriers, functions  $\Gamma_m(\square_1, \square_2, \square_3)$ , branch and permutation data, and all cubic coefficients remain open. In particular, unrestricted local integration by parts cannot be used across independently labelled  $\square_i$  [19, 18].

**Proposition 6.7** (Parity-even third-curvature carrier manifest). *On the declared asymptotically flat scalar-flat Euclidean conformal representative, set*

$$K_{\mu\nu} = \frac{2}{\square} \nabla^\beta \nabla^\alpha C_{\alpha\mu\beta\nu}.$$

The parity-even pure-gravity third-curvature sector has five carrier-labelled functions, denoted  $I_{10}, I_{24}, I_{25}, I_{28}, I_{29}$ , in the covariant-perturbation-theory basis. Their schematic tensor rows and label stabilizers are

|          |                                                         |                                      |
|----------|---------------------------------------------------------|--------------------------------------|
| $I_{10}$ | $\text{tr}(K_1 K_2 K_3)$                                | $S_3$                                |
| $I_{24}$ | $K_1 \nabla K_2 \nabla K_3$                             | $S_2(23)$                            |
| $I_{25}$ | $K_1 \nabla K_2 \nabla K_3$ (crossed)                   | $S_2(23)$                            |
| $I_{28}$ | $\nabla K_1 \nabla K_2 \nabla \nabla K_3$               | $S_2(12)$                            |
| $I_{29}$ | $\nabla \nabla K_1 \nabla \nabla K_2 \nabla \nabla K_3$ | $C_3$ (source), $S_3$ (scalar-flat). |

The last enhancement is not assumed from the source notation. In Fourier space the transverse scalar-flat carrier factorizes as

$$I_{29}(1, 2, 3) = -(k_2 K_1 k_2)(k_3 K_2 k_3)(k_1 K_3 k_1).$$

Momentum conservation and  $k_i K_i = 0$  identify it with  $I_{29}(1, 3, 2)$ . Thus the effective scalar-flat label modules have dimensions 1, 3, 3, 3, 1, with total dimension eleven and  $S_3$ -character  $(11, 5, 2)$ , or  $5\mathbf{1} \oplus 3\mathbf{2}_{\text{std}}$ . The source-generic  $C_3$  stabilizer remains recorded separately.

In four dimensions there is one integrated identity for every completely symmetric  $F(\square_1, \square_2, \square_3)$ . Writing  $x_i = \square_i$ , its pure- $K$  part is proportional to

$$-\frac{1}{12}(x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_2x_3 - 2x_1x_3)I_{10} - \frac{1}{2}x_1I_{24} - \frac{1}{2}(x_2 + x_3 - x_1)I_{25} + I_{28},$$

modulo total derivatives and  $O(\mathcal{R}^4)$ ;  $I_{29}$  is absent. Removing the fully symmetric component of  $I_{28}$  therefore leaves a ten-dimensional effective scalar-flat label module with character  $(10, 4, 1)$ , or

$$4\mathbf{1} \oplus 3\mathbf{2}_{\text{std}}.$$

The  $I_{29}$  row is the ineliminable lineage of the local parity-even algebraic  $C^3$  carrier, but its normalization in the two conventions is not inferred [19, 21].

This is a EUCLIDEAN-SPECTRAL carrier and permutation- quotient theorem. It is five carrier-labelled functions, with the effective  $I_{29}$  symmetry and one functional relation, not ten computed form factors. The five repository functions, all their coefficients, and the parity-odd derivative-decorated carrier manifest remain open.

**Proposition 6.8** (Universal CPT kernels and the repository matching boundary). *For the rank-one minimal scalar-Laplacian fixture  $F = -\square, P = 0, \mathcal{R}_{\mu\nu} = 0$ , the five source functions associated with  $I_{10}, I_{24}, I_{25}, I_{28}, I_{29}$  are imported exactly in the alpha-parameter convention*

$$\Gamma_i = \left\langle \frac{df_i}{-\Omega} \right\rangle_3 + T_i + L_i, \quad \Omega = \alpha_2 \alpha_3 \square_1 + \alpha_1 \alpha_3 \square_2 + \alpha_1 \alpha_2 \square_3.$$

Their stabilizers and box homogeneities are

| $i$                                | 10    | 24        | 25        | 28        | 29    |
|------------------------------------|-------|-----------|-----------|-----------|-------|
| $\text{Stab}_{\text{source}}(I_i)$ | $S_3$ | $S_2(23)$ | $S_2(23)$ | $S_2(12)$ | $C_3$ |
| $\text{Stab}_{K,\text{sf}}(I_i)$   | $S_3$ | $S_2(23)$ | $S_2(23)$ | $S_2(12)$ | $S_3$ |
| $\deg_{\square} \Gamma_i$          | -1    | -2        | -2        | -3        | -4.   |

The exact source formulas, symmetrized rows and formula hashes are retained in the computational receipt [19].

This imports universal kernels, not the pure-Weyl conformal-graviton functions. For tensor and ghost Laplace-type blocks, both the endomorphism  $P$  and bundle curvature  $\mathcal{R}_{\mu\nu}$  are already linear in the spacetime curvature and feed the same pure-gravity carriers through other CPT rows. The available round-sphere constrained determinant factors, Ricci-flat local  $b_4$  match and principal-symbol sequence do not determine those generic-background trace substitutions. Consequently no repository third-curvature function or coefficient is inferred by summing bundle ranks. The minimal missing physical input was therefore a same-gauge generic-background metric Hessian. The next propositions close its complete linear-curvature layer and import its projected algebraic curvature-squared block, while retaining the mixed trace as an explicit open edge.

**Proposition 6.9** (Same-gauge physical-Hessian linear-curvature import). *In the repository vector gauge*

$$F_\mu = \nabla^\nu h_{\mu\nu} - \frac{1}{4} \nabla_\mu h, \quad F_W = h,$$

the gauge-fixed traceless metric Hessian of Barvinsky et al. [8] is imported through first curvature order as

$$\mathcal{H}_{\text{src}} = \widehat{1}\square^2 + V^{\rho\sigma} \nabla_\rho \nabla_\sigma + N^\rho \nabla_\rho + U + O(\mathcal{R}^2).$$

The source gauge parameters  $(\gamma_1, \gamma_2, \tau) = (1/2, -1/6, -1/4)$  reproduce this repository gauge after imposing  $h = 0$ . The traceless projector has pointwise rank nine. The exact source ledger contains respectively nine, eight and five  $V, N, U$  rows; on the scalar-flat domain seven, six and three survive. Their rational coefficient vectors, index seeds, symmetrization policy and source hashes are stored in the machine receipt.

Because the source writes  $S^{(2)} = (1/4) \int \sqrt{g} h \mathcal{H}_{\text{src}} h$ , the repository functional Hessian is

$$\mathcal{H}_{\text{repo}} = \frac{1}{2} \mathcal{H}_{\text{src}}.$$

This agrees with the repository flat-TT leading coefficient. The constant factor cancels from the normalized insertion,

$$((\mathcal{H}_0/2)^{-1})(\mathcal{H}_1/2) = \mathcal{H}_0^{-1} \mathcal{H}_1.$$

On round  $S^4$ , writing  $A = -\square$  and  $K$  for sectional curvature, the imported source rows give  $A^2 + 6KA$ , while the certified complete repository fixture is

$$\frac{1}{2}(A + 2K)(A + 4K) = \frac{1}{2}A^2 + 3KA + 4K^2.$$

Thus the linear check passes and the missing  $+8K^2$  source term visibly certifies that this is not the full generic Hessian.

The import completely fixes the three-linear vertex

$$\text{Tr}[(\mathcal{H}_0^{-1}\mathcal{H}_1)^3].$$

The covariant rows have now also been converted to the exact scalar-flat rank-nine momentum vertex. Formal adjunction requires

$$A(k, q) = \frac{1}{2}[S(k, q) + S(-k, q + k)^T], \quad A(k, q) = A(-k, q + k)^T.$$

The completed vertex has zero adjoint defects; the uncompleted source seed has 62 on the stored rational check. At one generic nonexceptional fixture,

$$(k_1^2, k_2^2, k_3^2) = (10, 9, 5), \quad (\alpha_0, \alpha_1, \alpha_2) = \left(\frac{7}{15}, \frac{1}{5}, \frac{1}{3}\right), \quad \Delta = \frac{104}{45},$$

the exact loop trace has 210 monomials through loop degree six. Its four Wick orders give, before the universal  $(4\pi)^{-2}$  factor,

$$-\frac{3532544138843839}{319810083840000} \neq 0.$$

The full alpha polynomial has now been computed in an exact sparse algebra and projected onto all eleven raw channels of the ten-dimensional scalar-flat five-carrier quotient. The scalar-flat  $TT$  reconstruction of one linearized Riemann tensor on each leg introduces the uniform coordinate denominator  $x_1x_2x_3$ . After clearing it, a carrier with  $d$  explicit derivatives has numerator degree  $6 - d/2$ . Twenty-eight deterministic box fixtures give full degree-six evaluation rank modulo 1000003, hence full rank over  $\mathbb{Q}$ ; two additional unseen exact momentum fixtures have zero defects in every channel. The resulting 5755-term formula has digest `7c 91 74 4d a0 94 93 9e 86 e9 6a 68 07 b8 f4 06 c0 1b 1e 4b 29 91 ef 23 5e 11 de a2 c2 b3 b3 fe`. The common inverse boxes belong to the declared  $K/\text{Ricci}$  carrier coordinate and are not asserted to be new physical poles.

This closes the full-alpha and five-carrier-projection subgate. It does not integrate the generic tensor triangle, compute the mixed  $\mathcal{H}_1\mathcal{H}_2$  rows, or assemble the five complete repository form factors. The algebraic  $\mathcal{H}_2$  input is supplied next.

**Proposition 6.10** (Physical algebraic curvature-squared Hessian import). *For the linear background split, Ohta and Percacci [9] print the projected monic zero-order tensor  $U$  of the quadratic-curvature Hessian. Specializing their couplings to*

$$(\alpha, \beta, \gamma) = \left(\frac{1}{6}, -1, \frac{1}{2}\right), \quad \alpha R^2 + \beta R_{\mu\nu}^2 + \gamma R_{\mu\nu\rho\sigma}^2 = \frac{1}{2}C^2,$$

gives  $K = 1/4$  on the traceless bundle and hence monic  $\mathcal{H}_2 = 4PWP$ . The exact ledger retains all eighteen projected rows, including six trace-null rows and the source cancellation  $(\beta/2 + \gamma)R_{\mu\alpha}R_{\nu\beta} = 0$ . Nine nonzero rows survive on the scalar-flat domain.

The source orders the auxiliary gauge operator with  $(c, d) = (2/3, 1)$ , whereas the repository operator corresponds to  $(c, d) = (-1/3, 0)$  at the common constraint  $c - d = -1/3$ . Their difference is not an untracked change of  $\mathcal{H}_2$ :

$$Y_d - Y_0 = d[\nabla_\mu, \nabla_\nu], \quad S_{\text{gf}}(0) - S_{\text{gf}}(1) = \frac{1}{2} \int \sqrt{g} \chi_\mu R^{\mu\nu} \chi_\nu, \quad \chi_\mu = \nabla^\nu h_{\mu\nu}.$$

In monic-Hessian normalization this is the exact linear-curvature carrier

$$\mathcal{H}_{1,\text{repo}} - \mathcal{H}_{1,\text{src}} = 2G_{\text{Ric}}, \quad G_{\text{Ric}}(L, H) = (\nabla \cdot L)_\mu R^{\mu\nu} (\nabla \cdot H)_\nu.$$

It contains one background Ricci tensor and derivatives on both fluctuations, so it changes  $\mathcal{H}_1$  and contributes zero to the algebraic curvature-squared  $\mathcal{H}_2$  block.

On round  $S^4$  the thirteen source- $W$  contributions sum exactly to  $+24K^2$  on  $TT$  tensors. This is not the full order- $K^2$  remainder: contracted derivative indices in  $\mathcal{H}_1$  contribute  $-16K^2$ , and

$$24K^2 - 16K^2 = 8K^2$$

reproduces the monic factorization  $(A + 2K)(A + 4K)$ . This closes the algebraic  $\mathcal{H}_2$  input. Its operational polarization and first exact mixed fixture are supplied next.

**Proposition 6.11** (Raw mixed physical corner/endpoint fixture). *On the exact scalar-flat  $TT$  fixture*

$$k_1 = (1, 0, 0, 0), \quad k_2 = \frac{1}{2}(-1, 1, 1, 1), \quad k_3 = \frac{1}{2}(-1, -1, -1, -1), \quad k_i^2 = 1,$$

the nine nonzero  $\mathcal{H}_2$  rows admit an exact polarization between each pair of labelled external curvatures. All three polarized matrices have rank nine and are formally self-adjoint. The six labelled  $\mathcal{H}_1^3$  orderings and both endpoints of all three labelled  $\mathcal{H}_1\mathcal{H}_2$  bubbles give, before the common  $(4\pi)^{-2}$  factor,

$$c_{\mathcal{H}_1^3}^{\text{raw}} = -\frac{1975}{72}, \quad c_{\mathcal{H}_1\mathcal{H}_2}^{\text{raw}} = \frac{2704}{27}, \quad c_{\text{physical}}^{\text{raw}} = \frac{15707}{216} \neq 0.$$

The two terms live on different Feynman-parameter carriers: the first is a triangle-corner logarithm and the second a bubble-endpoint logarithm. Thus the mixed term cannot be added to the interior triangle numerator.

This exact nonzero fixture refutes a universal algebraic identity in which  $\mathcal{H}_2$  automatically cancels the isolated  $\mathcal{H}_1$  corner. It does not yet dispose the  $M_{14}$  relative class or define a renormalized physical coefficient. Those claims require a fixed local covariant subtraction/distribution extension that identifies the corner and endpoint carriers, followed by the renormalized mixed rows.

On this fixture the distribution extension is now fixed explicitly. Each labelled triangle is split into its three largest-barycentric sectors, each bubble into left and right half-intervals, and all boundary charts use the same Mellin parameter and  $z = \mu^2/Q^2$ . Minimal subtraction of  $z^s/s = 1/s + \log z + O(s)$  therefore gives the exact scale row

$$\frac{\partial \Gamma_{\text{fixture}}^{\text{MS}}}{\partial \log \mu^2} = \frac{1}{(4\pi)^2} \frac{15707}{216}.$$

This promotes the raw coefficient to a scheme-independent scale coefficient under  $\mu$ -independent finite local redefinitions on the declared fixture.

The generic covariant carrier of this subtraction is also exact. If  $H_0 = L^2$  is the monic principal operator, then

$$G_0 = H_0^{-1} = \int_0^\infty u e^{-uL} du.$$

Polarization gives six ordered three- $H_1$  Volterra cells and three local  $H_1$ - $H_2$  contact cells. In a triangle corner  $(u_1, u_2, u_3) = T(1-r, rt, r(1-t))$ , with Jacobian  $T^2 r$  and squared-propagator measure  $T^5 r^3(1-r)t(1-t)$ ; a contact cell has  $(u_1, u_2) = T(x, 1-x)$  and measure  $T^3 x(1-x)$ . The same Mellin extension is defined on all resolved boundaries. The generic logarithmic endpoint residues are now exact. For singled leg  $i$  and paired legs  $j, k$ ,

$$\text{Res}_{i,0} = -\frac{\text{tr}(H_{2,jk}H_{1,i}(k_i, 0))}{2(k_i^2)^2}, \quad \text{Res}_{i,1} = -\frac{\text{tr}(H_{2,jk}H_{1,i}(k_i, -k_i))}{2(k_i^2)^2},$$

and the two residues agree exactly. All three contacts project to 33 exact raw five-carrier rows, with two unseen momentum replays and the symmetric  $I_{28}$  quotient relation checked coefficientwise. Their scale kernels are  $2\text{Res}_{i,0} \log(\mu^2/x_i)$ . At the symmetric point the full six-ordering triangle row evaluates to  $-1975/72$  and the six contact endpoints to  $2704/27$ , hence their exact incidence is the nonzero scale row  $15707/216$ . Algebraic  $H_2$  cancellation of the symmetric  $M_{14}$  divergence is therefore refuted. The three generic-box triangle corner residues are now exact rational functions for all eleven channels. Their full incidence with the contact rows is nonzero, so generic algebraic  $H_2$  cancellation is refuted and  $M_{14}$  is disposed as a Mellin-renormalized scale row. The finite contact rows are also exact in the declared subtraction. Writing the loop-independent contact numerator as

$$C(t) = C(-p)(1-t) + C(0)t + c_2 t(1-t), \quad c_2 = 4C(-p/2) - 2C(-p) - 2C(0),$$

and denoting the Wick-pair coefficient by  $P$ , common Mellin minimal subtraction leaves

$$\text{FP}_{s=0} H_1 H_2 = -\frac{c_2}{2p^4} - \frac{P}{4p^2}.$$

All three contacts therefore project to 33 exact homogeneous rational rows, with two unseen tensor replays and coefficientwise  $I_{28} = 0$ . Their equal-box  $TT$  sum is  $3188/27$  before  $(4\pi)^{-2}$ . This fixes the selected contact finite part, not an arbitrary  $\mu$ -independent finite counterterm. The remaining triangle carrier problem is now exact. The 46 canonical relative-IBP tangent columns and three scalar-triangle masters have rank 49; adjoining

$$e_3, \quad e_3(\alpha_1 - \alpha_2), \quad e_3(\alpha_2 - \alpha_0)$$

raises the generic ranks successively to 50, 51, 52. The last two carriers form the standard two-dimensional  $S_3$  representation. Five exact orbit- representative rank tests and the generated  $S_3$  crosswalk place all eleven physical rows in this span. The three new renormalized master values are now explicit sector-decomposed rational/logarithmic functions in the common Mellin scheme, with exact scale derivatives. All sixty-six reduced rational master-coordinate functions of the eleven physical channels are also exact. Their relative-IBP boundary fluxes are now exact. The 33 corner numerators are at most quadratic in the angular parameter and reduce to the moments

$$\frac{a+b}{2a^2b^2}, \quad \frac{1}{2ab^2}, \quad \frac{\log(b/a) + 2a/b - a^2/(2b^2) - 3/2}{(b-a)^3}.$$

Thus all eleven channels have complete structured decompositions in  $J_\Delta$ , two bubble-log ratios, one rational corner term, and the  $M_{14}, M_{15}, M_{16}$  masters. Exact holdouts independently replay the tangent primitive, corner integrations and scale recipes, and all eleven symmetric scale rows reproduce the earlier obstruction ledger. Assembly with the finite contact rows is now exact. The eleven triangle channels, three finite contact rows per orientation and combined scale rows define five carrier-labelled physical-Hessian form-factor functions in the declared Mellin-MS representative.

Their orientation multiplicities are  $1 + 3 + 3 + 3 + 1$ , and the symmetric  $I_{28}$  component vanishes coefficientwise, leaving the ten-dimensional quotient. This does not yet give the complete repository form factors: the independent finite local  $C^2$  normalization and the ghost and remaining BV rows remain open. The complete integrated ghost  $n = 3$  contribution is added below.

**Proposition 6.12** (Generic ghost obstruction to direct minimal-CPT matching). *For the covariant gauge rows*

$$\delta F_\mu = \square \xi_\mu + R_{\mu\nu} \xi^\nu + (1 - 2\beta) \nabla_\mu \nabla \cdot \xi + 2(1 - 4\beta) \nabla_\mu \omega, \quad \delta F_W = 2 \nabla \cdot \xi + 8\omega,$$

the algebraic Weyl-ghost Schur complement is independent of  $\beta$ :

$$(M_{\text{eff}} \xi)_\mu = \square \xi_\mu + R_{\mu\nu} \xi^\nu + \frac{1}{2} \nabla_\mu \nabla \cdot \xi.$$

At a unit covector its principal eigenvalues are  $(3/2, 1, 1, 1)$ , so the operator is elliptic but not of Laplace type. No fixed invertible left and right bundle maps can scalarize this symbol for all covectors: for two orthogonal unit covectors the exact comparison spectrum is  $(2/3, 3/2, 1, 1)$ . Moreover,

$$M_{\text{eff}}(\nabla c) = -\frac{3}{2} \nabla \Delta_0 c + 2 R_{\mu\nu} \nabla^\nu c,$$

so generic tracefree Ricci curvature sends a longitudinal input into a transverse component. The Hodge determinant split used by the Einstein four-factor ledger therefore does not extend covariantly. On an Einstein background it reduces exactly to

$$M_{\text{eff}}(\nabla c) = -\frac{3}{2} \nabla (\Delta_0 - R/3) c,$$

recovering the accepted scalar ghost factor.

Consequently the imported minimal-Laplace CPT kernels cannot be applied to the generic repository ghost sector by direct rank or endomorphism substitution. The missing coefficient-bearing input is a matched nonminimal-vector determinant/CPT calculation, or an exact local extension with its determinant and Jacobian proved equivalent, together with the generic physical fourth-order Hessian kernel. This is an obstruction to the current matching architecture, not a new anomaly, QME, or Lorentzian no-go theorem.

**Proposition 6.13** (Exact Endo–Duhamel reduction of the ghost determinant). *With the positive Euclidean convention  $H = -M_{\text{eff}}$ , set*

$$F = -\square \mathbf{1} + \text{Ric}, \quad H_0 = F - \frac{1}{2} \nabla \nabla \cdot, \quad W = -2 \text{Ric}.$$

Then  $H = H_0 + W$  exactly. The base  $H_0$  is the nondegenerate Endo vector operator with parameter  $\alpha = -1/2$ , whose transverse and longitudinal principal eigenvalues are 1 and  $3/2$ . The Ward identities for  $F$  give the exact heat kernel [20]

$$K_{H_0}(t) = K_F(t) - \nabla \nabla' \int_t^{3t/2} ds K_{\Delta_0}(s).$$

Thus, on nonzero modes of a compact background without boundary,

$$\det' H_0 = \det' F \frac{\det'((3/2)\Delta_0)}{\det' \Delta_0},$$

and the last ratio contributes only the local zeta-scaling term  $\zeta_{\Delta_0}(0) \log(3/2)$ .

Since  $W = O(\mathcal{R})$ , the Duhamel expansion through  $O(\mathcal{R}^3)$  terminates, at the declared order, after the one-, two- and three- $W$  insertion traces. More precisely, the zero-, one-, two- and three-insertion rows require the Endo kernels only through curvature orders 3, 2, 1, 0, respectively. This converts the missing generic ghost calculation into a finite coefficient problem. The insertion traces and their carrier coordinates are not yet complete. The computational supplement now reduces the generic nonexceptional-momentum three-insertion triangle to eight exact projector sectors and twenty rational simplex/Wick rows. On the declared scalar-flat inverse domain, the contracted Weyl and Bianchi identities independently fix

$$K_{\mu\nu} = \frac{2}{\square} \nabla^\beta \nabla^\alpha C_{\alpha\mu\beta\nu} = R_{\mu\nu} + O(\mathcal{R}^2),$$

so the cubic replacement of three labelled  $K$  tensors by three labelled Ricci tensors is exact modulo  $O(\mathcal{R}^4)$ . A direct flat-TT tensor fixture fixes the sign. This closes the carrier normalization needed by the tensor projection: the zero-derivative sector can feed  $I_{10}$  and the longitudinal sectors can feed  $I_{24}, I_{25}, I_{28}, I_{29}$ . The next proposition performs that projection. The curved-Endo one-/two-insertion rows are reduced to minimal resolvent carriers below; their evaluation and the renormalized mixed physical-Hessian trace remain separate open calculations. The exact raw equal-box fixture below does not replace the missing covariant subtraction.

**Proposition 6.14** (Exact Hodge-resolvent reduction of the ghost  $n = 1, 2$  rows). *On primed nonzero modes, integration of the exact Endo heat-kernel identity gives*

$$G_{H_0} = G_F - \frac{1}{3}L, \quad L = d\Delta_0^{-2}\delta.$$

Indeed, after exchanging the proper-time integrations the domain is  $2s/3 \leq t \leq s$ , of width  $s/3$ , and  $\int_0^\infty sK_{\Delta_0}(s)ds = \Delta_0^{-2}$ . Writing  $W = -2\text{Ric}$ , cyclicity therefore reduces the complete one- and two-insertion nonminimal architecture to the following five minimal vector/scalar carriers:

$$\begin{aligned} \Gamma_{mgh}^{(1)} &= \text{Tr}_1(G_F W) - \frac{1}{3} \text{Tr}_0(\Delta_0^{-2} \delta W d), \\ \Gamma_{mgh}^{(2)} &= -\frac{1}{2} \text{Tr}_1(G_F W G_F W) + \frac{1}{3} \text{Tr}_0(\Delta_0^{-2} \delta W G_F W d) \\ &\quad - \frac{1}{18} \text{Tr}_0(\Delta_0^{-2} \delta W d \Delta_0^{-2} \delta W d). \end{aligned}$$

The coefficients  $(1, -1/3)$  and  $(-1/2, 1/3, -1/18)$  are exact. An independent rational fixture uses a symmetric, noncommuting  $W$  that mixes the transverse and longitudinal sectors and reproduces the direct resolvent traces through  $n = 3$ .

This closes the nonminimal reduction. The next proposition evaluates the combined pure-vector slice; the three carriers containing  $\delta W d$  remain outside the imported minimal-potential kernels.

**Proposition 6.15** (Pure-vector  $n = 1 + n = 2$  CPT projection and minimal missing-carrier theorem). *On the scalar-flat source complement, the positive minimal vector operators  $F = -\square + \text{Ric}$  and  $F + W = -\square - \text{Ric}$  correspond in the CPT-IV convention to  $P_F = -\text{Ric}$  and  $P_H = +\text{Ric}$ . Even- $P$  rows cancel and single- $P$  scalar traces vanish. The combined physical pure-vector contribution is therefore exactly*

$$6\Gamma_1 \mathcal{S}_1 - 2\Gamma_3 \mathcal{S}_3 - 2\Gamma_{14} \mathcal{S}_{14},$$

where  $\Gamma_1, \Gamma_3, \Gamma_{14}$  are the hash-pinned CPT-IV rows 1, 3 and 14. In the ordered channel basis

$$(I_{10}; I_{24}^{123}, I_{24}^{213}, I_{24}^{312}, I_{25}^{123}, I_{25}^{213}, I_{25}^{312}, I_{28}^{123}, I_{28}^{132}, I_{28}^{231}, I_{29}),$$



the two nontrivial source structures project as

$$\begin{aligned}\mathcal{S}_3 &\mapsto \left(-2, -\frac{1}{x_2}, -\frac{1}{x_1}, -\frac{x_1+x_2}{x_1x_2}, -\frac{2}{x_1}, -\frac{2}{x_2}, 0, 0, 0, 0, 0\right), \\ \mathcal{S}_{14} &\mapsto \left(-\frac{x_1+x_2-x_3}{2}, 0, 0, -1, -1, -1, 0, 0, 0, 0, 0\right).\end{aligned}$$

Ten exact momentum fixtures and all  $5^3$  transverse-tracefree products verify 2,500 projection identities; 750 further rows fix the Riemann/Ricci and contracted-Bianchi convention. An independent unseen-fixture replay rejects a coordinate mutation.

The remaining three Hodge carriers contain

$$D_W = \delta W d, \quad \sigma_2(D_W)(p) = W^{\mu\nu} p_\mu p_\nu.$$

This is a curvature-dependent anisotropic principal-symbol insertion, not a minimal bundle potential  $P$ ; the mixed carrier also couples scalar and vector resolvents. Hence the imported minimal-Laplace CPT rows do not evaluate these three carriers separately. The next proposition resums them into one scalar Schur determinant; its generic kernel, the physical fourth-order Hessian, and the complete repository functions remain open.

**Proposition 6.16** (Exact longitudinal ghost Schur resummation). *On the same primed nonzero-mode domain, write*

$$A = F + W, \quad H = A + \frac{1}{2}d\delta, \quad H_0 = F + \frac{1}{2}d\delta.$$

The Ward identities  $Fd = d\Delta_0$ ,  $\delta F = \Delta_0\delta$ , and  $\delta d = \Delta_0$  give the normalized scalar Schur operator

$$S_L(W) = \frac{2}{3}\mathbf{1} + \frac{1}{3}\delta(F+W)^{-1}d, \quad \sigma_0(S_L) = 1.$$

For one compatible determinant-class relative prescription, the matrix determinant lemma gives

$$\text{Det}_{\text{rel}}(H, H_0) = \text{Det}_{\text{rel}}(F+W, F) \text{Det}_F(S_L(W), \mathbf{1}).$$

If

$$B_n = \Delta_0^{-1} \delta W (G_F W)^{n-1} d \Delta_0^{-1},$$

then  $S_L = \mathbf{1} + \sum_{n \geq 1} (-1)^n B_n / 3$ , and its logarithm begins

$$\begin{aligned}O(W) &: -\frac{1}{3} \text{Tr } B_1, \\ O(W^2) &: \frac{1}{3} \text{Tr } B_2 - \frac{1}{18} \text{Tr } B_1^2, \\ O(W^3) &: -\frac{1}{3} \text{Tr } B_3 + \frac{1}{9} \text{Tr}(B_1 B_2) - \frac{1}{81} \text{Tr } B_1^3.\end{aligned}$$

The first two lines reproduce the three previously isolated longitudinal carriers, and the third fixes their complete cubic weights. A distinct noncommuting exact fixture verifies the determinant and trace-log identities; on an Einstein background the minimal-vector longitudinal factor times  $S_L$  reproduces  $(\Delta_0 - R/3)/\Delta_0$ .

This is a resummation theorem, not a form-factor evaluation. A separately zeta-regularized factorization can carry a local multiplicative anomaly. Both a commuting round- $S^4$  ratio convention

and the canonical generic weight-raised convention  $A = S_L Q$ ,  $B = Q$  are computed below. They must not be identified: a multiplicative defect depends on the chosen factorization. Moreover  $S_L - \mathbf{1}$  starts at pseudodifferential order  $-2$  in four dimensions, so ordinary trace-class Fredholm determinacy is not implied by order counting. A regularized relative determinant or equivalent common trace regulator is the remaining global longitudinal coefficient input.

**Proposition 6.17** (Non-Einstein product Schur spectrum and coupled priming). *On  $S^2(k_1) \times S^2(k_2)$ ,  $k_i > 0$ , let*

$$a = k_1 \ell(\ell + 1), \quad b = k_2 m(m + 1), \quad \lambda = a + b, \quad d_{\ell m} = (2\ell + 1)(2m + 1).$$

*For each active factor gradient, the minimal-vector exact eigenvalue is  $\lambda - 2k_i$ , and the normalized Schur eigenvalue is*

$$s_{\ell m} = \frac{2}{3} + \frac{1}{3} \sum_{i \in \text{active}} \frac{a_i}{\lambda - 2k_i}.$$

*These formulas supply a complete compact non-Einstein spectral measure for the longitudinal kernel. They are covariant under  $(k_1, \ell) \leftrightarrow (k_2, m)$  and specialize at  $k_1 = k_2 = k$  to*

$$s_{\ell m} = \frac{\lambda - 4k/3}{\lambda - 2k} = \frac{\Delta_0 - R/3}{\Delta_0 - R/2}, \quad R = 4k.$$

*The six modes  $(1, 0)$  and  $(0, 1)$  require a coupled prescription. Their minimal-vector factor vanishes while  $s_{\ell m}$  has a pole, but the product has the finite continuation*

$$\frac{\lambda - 2k_i}{\lambda} s_{\ell m} \longrightarrow \frac{1}{3}.$$

*Thus the regular-complement determinant must be multiplied by  $3^{-6}$ ; priming the vector zeros and scalar rows independently loses a finite factor. An independent exact matrix replay verifies this correction and the finite cutoff products. The principal spectral symbol also reproduces*

$$\text{Wres}(K^2) = \frac{8(k_1^2 + k_1 k_2 + k_2^2)}{27k_1 k_2}.$$

*The weighted finite-part sums are not evaluated here, so this is a background-specific spectral carrier rather than a completed ghost determinant or generic form factor. The regular-complement  $\det_3$  sum on one non-Einstein fixture is evaluated next.*

**Proposition 6.18** (Product-background Schur  $\det_3$  enclosure). *On  $S^2(1) \times S^2(2)$ , remove the constant scalar mode and the two matched zero-pole levels  $(1, 0)$  and  $(0, 1)$ . For the remaining product harmonics, the canonical third modified determinant obeys*

$$0.32630391405410603404 \dots < \log \det_3(\mathbf{1} + K) < 0.32630396588976784255 \dots$$

*In particular, its certified common decimal prefix is 0.3263039.*

*The enclosure combines an exact rational alternating Taylor bound on the finitely many modes with  $K > 1/100$ , an ordinary positive finite sum with an explicit  $\gamma_N$  roundoff proof and Taylor-remainder cushions on the remaining rectangle, and an exact rational exterior-tail majorant. An independent replay uses a different stable order-twelve series. The exceptional coupled correction  $3^{-6}$  is retained separately: it is not part of this regular-complement modified determinant. The weighted  $\mathcal{R}(K)$  and finite-part  $\mathcal{R}(K^2)$  rows remain open, as do the full coupled ghost determinant and generic-background finite form factors.*

**Proposition 6.19** (Sharp Schatten split and critical Schur residue). *Set  $K = S_L - \mathbf{1}$  on a closed compact four-manifold, with the primed zero-mode complement and elliptic inverses fixed. The exact resolvent identity gives*

$$K = -\frac{1}{3}\delta(F + W)^{-1}Wd\Delta_0^{-1}, \quad \sigma_{-2}(K)(x, \xi) = -\frac{1}{3}\frac{\langle \xi, W\xi \rangle}{|\xi|^4}.$$

*Hence  $K$  is a classical scalar pseudodifferential operator of order  $-2$ . Its singular values obey  $s_j(K) = O(j^{-1/2})$  [24], so  $K \in \mathcal{S}_p$  for every  $p > 2$ , in particular  $K \in \mathcal{S}_3$ ;  $K^3$  is trace class. The third modified Fredholm determinant [23]*

$$\det_3(\mathbf{1} + K) = \det \left[ (\mathbf{1} + K)e^{-K+K^2/2} \right]$$

*is therefore canonical. For a declared local trace extension  $\mathcal{R}$ , the remaining renormalized split is*

$$\log \text{Det}_{3,\mathcal{R}}(\mathbf{1} + K) = \mathcal{R}(K) - \frac{1}{2}\mathcal{R}(K^2) + \log \det_3(\mathbf{1} + K).$$

*The critical order-4 residues are already fixed. In the convention*

$$\text{Wres } P = (2\pi)^{-4} \int_M \int_{|\xi|=1} \sigma_{-4}(P) dS d\text{vol},$$

*the four-sphere moment gives*

$$\text{Wres}(K^2) = \frac{1}{(4\pi)^2} \int_M \frac{(\text{tr } W)^2 + 2 \text{tr}(W^2)}{108} = \frac{1}{(4\pi)^2} \int_M \frac{R^2 + 2R_{\mu\nu}R^{\mu\nu}}{27}.$$

*The residue of  $K$  itself needs one subprincipal row. Through residue order, the resolvent expansion is*

$$K = -\frac{1}{3}B_1 + \frac{1}{3}B_2 + O(\Psi^{-6}), \quad B_1 = \Delta_0^{-1}\delta W d\Delta_0^{-1}, \quad B_2 = \Delta_0^{-1}\delta W G_F W d\Delta_0^{-1}.$$

*Cyclicity and the mixed scalar heat kernel, with the standard heat/residue normalization [25, 26], give*

$$\text{Wres}(B_1) = \frac{1}{(4\pi)^2} \int_M \left[ -\frac{1}{3}W^{\mu\nu}R_{\mu\nu} + \frac{1}{6}(\text{tr } W)R \right], \quad \text{Wres}(B_2) = \frac{1}{(4\pi)^2} \int_M \frac{1}{2} \text{tr}(W^2).$$

*Thus, for the repository insertion  $W = -2 \text{Ric}$ ,*

$$\text{Wres}(K) = \frac{1}{(4\pi)^2} \int_M \frac{R^2 + 4R_{\mu\nu}R^{\mu\nu}}{9},$$

*and the logarithmic residue is*

$$\text{Wres log } S_L = \text{Wres}(K) - \frac{1}{2} \text{Wres}(K^2) = \frac{1}{(4\pi)^2} \int_M \frac{5R^2 + 22R_{\mu\nu}R^{\mu\nu}}{54}.$$

*On an Einstein background the exact ratio  $S_L = (\Delta_0 - R/3)(\Delta_0 - R/2)^{-1}$  independently gives  $\text{Wres}(K) = (4\pi)^{-2} \int 2R^2/9$ . A covariantly constant isotropic test  $W = w\mathbf{1}$  likewise reduces  $B_1$  to  $w\Delta_0^{-1}$ . These checks fix the mixed-kernel sign and normalization.*

*These equations compute the canonical local residues, not the full Schur determinant. The generic regulated finite value  $\mathcal{R}(K)$  and the finite part of  $\mathcal{R}(K^2)$  remain open. The selected generic weight-raised zeta multiplicative term is computed below from these residues and the order-3/-4 BCH filtration. Converting  $\text{Wres log } S_L$  into a pole or scale coefficient requires the order and normalization of a chosen reference operator; the next proposition makes that choice explicitly.*

**Proposition 6.20** (Order-two weighted-trace pole and scale row). *On the primed scalar space let*

$$Q = \Delta_0 + \Pi_0, \quad \text{ord } Q = 2, \quad Q_\mu = Q/\mu^2,$$

*where the smoothing projector only restores invertibility. Declare*

$$\mathcal{R}_\mu(A) = \text{FP}_{z=0} \text{TR}'[AQ_\mu^{-z}].$$

*The weighted-trace Laurent formula [27] gives*

$$\text{TR}'(AQ^{-z}) = \frac{\text{Wres}(A)}{2z} + \mathcal{R}_Q(A) + O(z).$$

*Since  $Q_\mu^{-z} = \mu^{2z}Q^{-z}$ , multiplication of the Laurent series fixes both normalizations:*

$$\mathcal{R}_{e^t\mu}(A) - \mathcal{R}_\mu(A) = t \text{Wres}(A).$$

*In particular,*

$$\text{Res}_{z=0} \text{TR}'[(\log S_L)Q^{-z}] = \frac{1}{(4\pi)^2} \int_M \frac{5R^2 + 22R_{\mu\nu}R^{\mu\nu}}{108},$$

*and*

$$\boxed{\frac{d}{d \log \mu} \log \text{Det}_{3,\mathcal{R}_\mu}(S_L) = \text{Wres} \log S_L = \frac{1}{(4\pi)^2} \int_M \frac{5R^2 + 22R_{\mu\nu}R^{\mu\nu}}{54}.}$$

*This closes the reference-specific pole and scale conversion. It does not fix the generic reference-scale constants  $\mathcal{R}_{\mu_0}(K)$  and  $\text{FP } \mathcal{R}_{\mu_0}(K^2)$ . For the separately declared weight-raised zeta prescription, the local noncommuting term is computed below [28, 4].*

**Proposition 6.21** (Exact round- $S^4$  finite Schur benchmark). *On the round unit four-sphere, delete the absent constant-gradient row and the five  $\ell = 1$  proper-conformal-Killing ghost zero modes. On the remaining  $\ell \geq 2$  carrier,*

$$\lambda_\ell = \ell(\ell + 3), \quad d_\ell = \frac{(2\ell + 3)(\ell + 2)(\ell + 1)}{6}, \quad K_\ell = \frac{2}{\lambda_\ell - 6}.$$

*At  $\mu_0 = 1$  in inverse-radius units, set*

$$u_\pm = \frac{7 \pm \sqrt{33}}{2}, \quad qquad \Psi = \psi(u_-) + \psi(u_+), \quad \Psi_1 = \psi^{(1)}(u_-) - \psi^{(1)}(u_+).$$

*Hurwitz-zeta continuation of the complete spectrum [29] and the same-order weight-change formula give*

$$\boxed{\mathcal{R}_{\Delta_0}(K) = -\frac{20}{9} - \frac{8}{3}\Psi,}$$

$$\boxed{\text{FP } \mathcal{R}_{\Delta_0}(K^2) = -\frac{2}{3}\Psi + \frac{16}{3\sqrt{33}}\Psi_1.}$$

*Numerically these are  $-3.0967576144286354\dots$  and  $2.7591028732128106\dots$ . Hence the finite low-order part of the modified-determinant split is*

$$\mathcal{R}_{\Delta_0}(K) - \frac{1}{2} \text{FP } \mathcal{R}_{\Delta_0}(K^2) = -4.4763090510350407\dots$$

The remaining canonical tail is absolutely convergent. Exact alternating-series bounds on the first nineteen modes, followed by a rational Euler–Maclaurin enclosure of the Hurwitz-zeta tail, give

$$\log \det {}_3(I + K) = 0.4981635654196290984312532999414818723861192934\dots,$$

with certified enclosure width below  $5.8 \times 10^{-48}$ . Hence, for this declared weight and reference scale,

$$\log \text{Det}_{3, \mathcal{R}_{\Delta_0}}(S_L) = -3.9781454856154116274753955548059869205821661933\dots$$

This is a special-background benchmark, not the generic finite row. Indeed,  $K \mapsto K + T$  for arbitrary finite-rank smoothing  $T$  preserves every homogeneous symbol and Wodzicki residue but shifts the two finite rows by  $\text{Tr } T$  and  $\text{Tr}(KT + TK + T^2)$ . Thus the generic values require the full primed Green kernel or equivalent spectral measure. The local comparison for this commuting round-sphere factorization is computed next, followed by a distinct generic weight-raised convention.

**Proposition 6.22** (Exact round- $S^4$  zeta-to-weighted factorization defect). *On the same primed  $\ell \geq 2$  scalar carrier set*

$$Q = \Delta_0, \quad A = \Delta_0 - 4, \quad B = \Delta_0 - 6, \quad S_L = AB^{-1}.$$

These operators commute. The zeta/weighted-determinant comparison of Paycha and Ouedraogo–Paycha [27, 4] therefore reduces in four dimensions to

$$\begin{aligned} m_Q(A, B) &= \log \det_{\zeta} A - \log \det_{\zeta} B - \text{tr}^Q \log(AB^{-1}) \\ &= -\frac{1}{4}(4^2 - 6^2) \text{Wres}(Q^{-2}). \end{aligned}$$

The scalar heat-kernel residue gives

$$\text{Wres}(Q^{-2}) = 2 \text{Res}_{s=2} \zeta_Q(s) = 2 \frac{\text{Vol}(S^4)}{(4\pi)^2} = \frac{1}{3},$$

so the exact local defect is

$$m_Q(A, B) = \frac{5}{3}.$$

Adding it to the weighted modified determinant above gives

$$\log \det_{\zeta}(\Delta_0 - 4) - \log \det_{\zeta}(\Delta_0 - 6) = -2.3114788189487449608087288881393202539\dots$$

An independent Hurwitz-zeta continuation on the complete primed spectrum reproduces the same number. This is special-background and prescription-specific. It does not by itself compute the generic noncommuting local term or supply the generic global finite rows. The next proposition fixes a different, canonical weight-raised comparison and makes the factorization dependence explicit.

**Proposition 6.23** (Generic weight-raised Schur zeta factorization). *On a closed compact four-manifold, keep the primed scalar carrier and the positive order-two weight*

$$Q = \Delta_0 + \Pi_0.$$

There is no canonical generic numerator/denominator factorization analogous to the Einstein identity above. Instead declare

$$A = S_L Q, \quad B = Q,$$

with this left-Schur ordering and one common Agmon sector on a connected admissible stratum. Define

$$m_Q^{\text{wr}}(S_L) = \log \det_{\zeta}(S_L Q) - \log \det_{\zeta} Q - \text{tr}^Q \log S_L.$$

Put  $X = \log S_L \in \Psi^{-2}$  and  $Y = \log Q$ . Through the four-dimensional residue order, the BCH series is

$$\log(S_L Q) - X - Y = \frac{1}{2}[X, Y] + \frac{1}{12}[Y, [Y, X]] \pmod{\Psi^{-5}}.$$

The displayed commutators have orders  $-3$  and  $-4$ ; the next independent term  $[X, [X, Y]]$  has order  $-6$ . The weighted trace defect identity

$$\text{tr}^Q[U, V] = -\frac{1}{\text{ord } Q} \text{Wres}(U[V, \log Q])$$

kills both surviving BCH commutators because one entry is  $Y = \log Q$ . Cross terms with  $X$  begin below residue order, while  $(\log(1 + K))^2 = K^2 \pmod{\Psi^{-6}}$ . The zeta/weighted comparison therefore gives the exact local result

$$m_Q^{\text{wr}}(S_L) = -\frac{1}{4} \text{Wres}(K^2) = -\frac{1}{(4\pi)^2} \int_M \frac{R^2 + 2R_{\mu\nu}R^{\mu\nu}}{108} \text{dvol}.$$

On the round unit sphere this convention gives

$$\text{Wres}(K^2) = \frac{4}{3}, \quad m_Q^{\text{wr}}(S_L) = -\frac{1}{3},$$

and hence

$$\log \det_{\zeta}(S_L \Delta_0) - \log \det_{\zeta} \Delta_0 = -4.3114788189487449608087288881393202539 \dots$$

An independent Hurwitz-zeta continuation of the primed scalar spectrum replays this value. The difference between this defect and the preceding  $5/3$  Einstein-ratio defect is exactly 2; the values are not contradictory because the factorizations are different. This closes the selected generic local BCH/residue term, not the global generic finite Schur rows or a full vector-block determinant factorization.

**Proposition 6.24** (Exact ghost  $n = 3$  scalar-flat five-carrier projection). *Evaluate the eleven effective labelled orientations*

$$I_{10}; \quad I_{24}^{\times 3}; \quad I_{25}^{\times 3}; \quad I_{28}^{\times 3}; \quad I_{29}$$

on the  $5^3 = 125$  products of exact transverse-tracefree polarizations at generic nonexceptional Euclidean momenta. The resulting  $125 \times 11$  matrix has exact rank ten. Appending the quotient section

$$\Gamma_{28}^{123} + \Gamma_{28}^{132} + \Gamma_{28}^{231} = 0$$

fixes the unique CPT-IV four-dimensional null direction. In that section, the complete three-insertion ghost triangle has exact channel coordinates

$$\Gamma_i(\alpha; x) = \frac{N_i(\alpha_1, \alpha_2; x_1, x_2, x_3)}{\Delta^4}, \quad \Delta = \alpha_0 \alpha_1 x_1 + \alpha_1 \alpha_2 x_2 + \alpha_2 \alpha_0 x_3,$$

where every coefficient of every numerator  $N_i$  is rational. If carrier  $i$  has  $d_i$  explicit derivatives,  $N_i$  is homogeneous of box degree  $3 - d_i/2$  and has alpha degree at most nine. The  $W = -2\text{Ric}$  and third-order trace-log multiplier  $-8/3$  is included; the common  $(4\pi)^{-2}$  loop prefactor is not.

Ten box-unisolvent momentum fixtures reconstruct the stored polynomials. Two unseen momentum fixtures and unseen rational simplex points independently reproduce all 125 triangle amplitudes with zero exact residual, and a rehashed coefficient mutation is rejected. This is the parametric  $n = 3$  ghost contribution. It is not the integrated five repository functions, the evaluated five minimal  $n = 1/n = 2$  resolvent carriers, the complete ghost determinant, or the physical fourth-order Hessian contribution.

**Proposition 6.25** (Generic barycentric denominator and boundary factorization). Put  $A = \alpha_1$ ,  $B = \alpha_2$ ,  $C = \alpha_0$  and

$$\Delta = CAx_1 + ABx_2 + BCx_3.$$

Ten of the eleven exact projected numerators contain one factor of  $\Delta$ . Their pole order therefore drops from four to three at generic positive nonexceptional kinematics. The sole pole-four row is

$$\frac{-16A^3B^3C^3/27}{\Delta^4} = I_{29}^{123}.$$

After homogeneous barycentric lifting, the edge orders are  $(0,0,0)$  for  $I_{10}$ , permutations of  $(1,2,1)$  for  $I_{24}$ ,  $(1,1,1)$  for  $I_{25}$ ,  $(2,2,2)$  for  $I_{28}$  and  $(3,3,3)$  for  $I_{29}$ . Hence  $I_{10}$  is the only raw row with a nonzero direct restriction to an open simplex edge. Every vertex integrability margin  $v - p + 2$  is positive; the minimum is one. Moreover the reduced  $I_{28}$  numerators obey

$$\tilde{N}_{28}^{123} + \tilde{N}_{28}^{132} + \tilde{N}_{28}^{231} = 0$$

pointwise, before integration.

This theorem fixes the generic denominator and direct-boundary structure. It does not yet construct rational relative-IBP primitives, verify their punctured-corner flux, or determine whether bubble/log terms generated by those primitives vanish. The following theorem supplies that reduction for the ten pole-three rows; the subsequent proposition closes the pole-four row.

**Proposition 6.26** (Generic pole-three relative-simplex IBP reduction). For each of the ten pole-three rows there are rational functions  $U, V, W$  of affine degree at most four such that

$$P = A(CU + BW), \quad Q = B(CV - AW)$$

has zero normal component on every open simplex edge and

$$N = \Delta(\partial_A P + \partial_B Q) - 2(P\partial_A \Delta + Q\partial_B \Delta) + c_J \Delta^2 + c_1 AC\Delta + c_2 AB\Delta.$$

Consequently every row reduces exactly to

$$J_\Delta = \int_\Delta \frac{1}{\Delta}, \quad -\partial_{x_1} J_\Delta, \quad -\partial_{x_2} J_\Delta.$$

Four explicit primitives cover all ten orientations by exact simplex permutations. In the degree-seven polynomial carrier the open-edge tangent image has rank 27 and adjoining the three masters raises it to 30; adjoining every target leaves rank 30.

Open-edge tangency is not zero corner flux. Imposing  $U_A = W_A = V_B = W_B = U_C = V_C = 0$  leaves a tangent-plus-master span of rank 26, whereas every target raises the rank to 27. The

certificate stores a normalized dual witness for each orbit representative at a rational fixture whose base rank equals the generic rank:

$$\lambda(B_{\text{corner-zero}}) = 0, \quad \lambda(N) = 1.$$

Thus punctured-corner flux is unavoidable in the declared exhaustive degree ansatz. This does not by itself imply a new logarithmic master: the following proposition evaluates the angular flux and the derivative masters explicitly.

**Proposition 6.27** (Complete generic functions for the ten pole-three rows). *Let*

$$\lambda = x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 - 2x_2x_3.$$

The scalar-triangle differential system is

$$\lambda \partial_{x_1} J_\Delta = (x_2 + x_3 - x_1) J_\Delta - \frac{x_1 + x_2 - x_3}{x_1} \log \frac{x_2}{x_1} - \frac{x_1 + x_3 - x_2}{x_1} \log \frac{x_3}{x_1},$$

and its two cyclic images. The certificate verifies all six  $S_3$  relabellings, Euler homogeneity and all three mixed-derivative identities. Consequently the two derivative masters above reduce to  $J_\Delta$  and the two independent bubble-log ratios  $\log(x_2/x_1), \log(x_3/x_1)$ .

For each of the four canonical relative-IBP primitives, the two coordinate vertices have zero leading pair. At the sole nonzero vertex one has  $U_C = V_C$ , and hence

$$\int_0^{\pi/2} \frac{U_C \cos^2 \theta + V_C \sin^2 \theta}{(x_1 \cos \theta + x_3 \sin \theta)^2} d\theta = \frac{U_C}{x_1 x_3}.$$

Including the stored boundary orientation gives the rational corner term  $-U_C/(x_1 x_3)$ , transported by the certified simplex permutations. Therefore every pole-three channel is now a complete exact function

$$A_J(x) J_\Delta + A_{21}(x) \log \frac{x_2}{x_1} + A_{31}(x) \log \frac{x_3}{x_1} + R_{\text{corner}}(x),$$

with rational-function coefficients. All ten rows regress exactly to the symmetric-point values below, and the integrated  $I_{28}$  relation vanishes separately in all four basis coordinates. The remaining pole-four row is handled separately because its primitive has a different denominator order.

**Proposition 6.28** (Complete generic pole-four  $I_{29}$  function). *The sole pole-four carrier is*

$$I_{29}^{123} = -\frac{16}{27} \int_{\Delta_2} \frac{(\alpha_0 \alpha_1 \alpha_2)^3}{\Delta^4}.$$

In the 55-dimensional numerator space, the exact pole-four tangent image has rank 46. Adjoining the three masters  $J_\Delta, -\partial_{x_1} J_\Delta, -\partial_{x_2} J_\Delta$  raises the rank to 49, while adjoining the target leaves it at 49:

$$\text{rank } B = 46, \quad \text{rank}(B, M) = 49, \quad \text{rank}(B, M, I_{29}) = 49.$$

Thus no new transcendental master is required. The three master coefficients have common denominator  $\lambda^5$  and numerator degrees 7, 8, 8. A 45-point exact unisolvent reconstruction and five independent holdouts are followed by a full 55-row symbolic relative-IBP identity; the interpolation is therefore not the proof certificate by itself.

The canonical primitive has linear corner numerators. The only required angular moments are rational, so after substitution of the scalar-triangle differential system the result again has the form

$$A_J(x) J_\Delta + A_{21}(x) \log \frac{x_2}{x_1} + A_{31}(x) \log \frac{x_3}{x_1} + R(x).$$



All six  $S_3$  relabellings vanish exactly. At the symmetric point,

$$I_{29}(1, 1, 1) = -\frac{496}{6561}J_\Delta + \frac{1160}{6561},$$

which agrees with the independent simplex integration below. Hence all eleven generic scalar-flat ghost  $n = 3$  functions are complete. This does not assemble the full ghost determinant, determine the five repository form factors, or supply the covariantly subtracted mixed physical-Hessian trace.

**Proposition 6.29** (Physical plus ghost- $n = 3$  carrier assembly). *In the common scalar-flat carrier quotient, channelwise addition of the eleven integrated ghost- $n = 3$  functions to the physical-Hessian Mellin-MS representative is exact. The upstream bosonic and fermionic determinant multipliers are already included, and the common factor  $(4\pi)^{-2}$  remains excluded. The ghost triangle contributes to  $J_\Delta$ , the two bubble logarithms and the rational corner coordinate; the three resolved-boundary Mellin masters and the scale row remain the physical-Hessian values. The relation*

$$I_{28}^{123} + I_{28}^{132} + I_{28}^{231} = 0$$

holds coefficientwise after addition. The resulting partial BV representative is ten-dimensional. This does not include curved-Endo ghost  $n = 1/n = 2$ , generic finite Schur rows, remaining BV sectors or the independent finite  $C^2$  normalization, and is not a complete  $\Gamma_1$  or  $Q_1$ .

**Proposition 6.30** (Exact vector- $n = 1 + n = 2$  integration and partial-BV assembly). *For the pure minimal-vector slice, the projected numerator degree is at most two. Five exact simplex boundary identities—the two first-moment  $\partial_{\alpha_i} \log \Delta$  identities, two weighted cross-boundary identities, and  $\int_{\Delta_2} \Delta/\Delta = 1/2$ —therefore determine every required moment. All eleven channels reduce to*

$$J_\Delta, \quad \log \frac{x_2}{x_1}, \quad \log \frac{x_3}{x_1}, \quad \mathbb{Q}(x_1, x_2, x_3),$$

so no new transcendental master is required. Six channels are nonzero and five vanish identically; an independent direct quadrature checks all six nonzero functions at a nonexceptional point.

Adding these rows to the physical-plus-ghost- $n = 3$  representative preserves the coefficientwise  $I_{28}$  relation and the ten-dimensional quotient. The result is an exact partial-BV five-carrier Mellin-MS representative containing the same-gauge physical Hessian, all ghost  $n = 3$  rows, and  $N1_{\text{vector}} + N2_{\text{vector}, \text{vector}}$ . The longitudinal/mixed  $D_W = \delta W d$  towers are one Schur kernel, whose exact non-Einstein product spectrum is supplied above. Its regular-complement  $\det_3$  value on  $S^2(1) \times S^2(2)$  is enclosed above, while its weighted finite-part rows, arbitrary-background finite Schur data, and the remaining BV sectors are excluded; this is not the complete ghost determinant,  $\Gamma_1$ , or  $Q_1$ .

**Proposition 6.31** (Exact symmetric-point ghost triangle fixture). *At the normalized nonexceptional Euclidean point  $x_1 = x_2 = x_3 = 1$ , put*

$$e_2 = \alpha_0 \alpha_1 + \alpha_1 \alpha_2 + \alpha_2 \alpha_0, \quad e_3 = \alpha_0 \alpha_1 \alpha_2, \quad J_\Delta = \int_{\Delta_2} \frac{d^2 \alpha}{e_2} = \frac{4}{\sqrt{3}} \text{Cl}_2\left(\frac{\pi}{3}\right).$$

The eleven projected ghost coordinates integrate exactly to

$$\begin{aligned}
I_{10}^{123} &= -\frac{8}{2187}(592J_{\Delta} + 55), \\
I_{24}^{123} &= I_{24}^{213} = I_{24}^{312} = \frac{32}{6561}(7J_{\Delta} - 17), \\
I_{25}^{123} &= I_{25}^{213} = I_{25}^{312} = \frac{20}{6561}(88J_{\Delta} - 179), \\
I_{28}^{123} &= I_{28}^{132} = I_{28}^{231} = 0, \\
I_{29}^{123} &= -\frac{8}{6561}(62J_{\Delta} - 145).
\end{aligned}$$

The proof symmetrizes each exact numerator under the simplex permutation group and reduces the result to four moments of  $e_2, e_3$ . Each moment is certified by a rational divergence vector field whose normal flux vanishes on the three open edges and whose stored vertex orders kill every punctured-corner flux. An independent one-dimensional quadrature replays the Clausen master and guards against the inconsistent logarithm branches produced by naive iterated symbolic integration. The upstream  $W = -2\text{Ric}$  and trace-log factor  $-8/3$  are included; the common  $(4\pi)^{-2}$  factor is not. This is an independent coefficient-bearing symmetric-point fixture for the generic formulas above, not the physical Hessian contribution or the assembled repository form factors.

**Proposition 6.32** (Exact local  $\square R$  scheme conversion). *In the zeta/proper-time prescription of Barvinsky et al. [8], the imported fourth heat-kernel rows combine with weights  $(2, -1, -2)$  to give*

$$b_{\square}^{\text{raw}} = \frac{7}{2} \log \frac{3}{2} - \frac{159}{80} < 0.$$

The project convention

$$\omega \square R = -\frac{1}{12}s(R^2) \pmod{d}$$

therefore fixes the one-loop strict-metric counterterm

$$z_R^{\text{raw} \rightarrow 0} = \frac{b_{\square}^{\text{raw}}}{12} = \frac{7}{24} \log \frac{3}{2} - \frac{53}{320}.$$

It converts the raw coefficient to the repository  $\square R = 0$  convention and, when added to the raw anomaly-induced local  $R^2$  coefficient  $391/960 - (7/24)\log(3/2)$ , gives exactly  $29/120$ . This is a relative local one-loop scheme conversion. By itself it neither proves the preceding FV Ricci-sector dependence theorem nor fixes the independent dressed  $R(\hat{g})^2$  counterterm in the compensator theory, and it is not an all-loop equivalence theorem for strict Weyl gravity.

**Proposition 6.33** (Isolated physical three- $\mathcal{H}_1$  corner obstruction). *At  $x_1 = x_2 = x_3 = 1$ , let*

$$e_2 = \alpha_0\alpha_1 + \alpha_1\alpha_2 + \alpha_2\alpha_0, \quad e_3 = \alpha_0\alpha_1\alpha_2.$$

The  $S_3$  average of the exact five-carrier projection of  $\frac{1}{6} \text{Tr}[(\mathcal{H}_0^{-1}\mathcal{H}_1)^3]$  contains the pole-four carrier  $M_{14} = e_3/e_2^4$ . In the declared 55-dimensional affine numerator space, the exact relative-IBP-plus-master span has rank 49, while adjoining  $e_3$  raises the rank to 50. A stored dual functional annihilates the full rank-49 span and is normalized to one on  $e_3$ .

Near each simplex corner the measure-weighted carrier is

$$\frac{t(1-t)}{\epsilon} d\epsilon dt + O(1),$$

*so its logarithmic coefficient is  $1/6$  per corner and  $1/2$  in total. Eight raw orientations have a nonzero  $M_{14}$  coefficient:  $I_{10}$ , all three  $I_{24}$  rows, all three  $I_{25}$  rows, and  $I_{29}$ ; all three  $I_{28}$  rows have zero coefficient in this obstruction coordinate. Therefore the isolated three- $\mathcal{H}_1$  trace is not a finite physical simplex integral. The imported curvature-squared  $\mathcal{H}_2$  must now be polarized and its mixed rows tested against the same class, or a renormalized subtraction must be fixed. No cancellation by  $\mathcal{H}_2$  is asserted.*

## 7 What does not follow

The strict obstruction and coefficient-bearing extended restoration are local one-loop results; the all-order extension above is a conditional formal local induction. They do not establish:

- an unconditional all-loop extended QME or a regulator realizing the declared quantum action principle;
- a Lorentzian renormalized time-ordered product or Slavnov identity;
- a BRST-compatible global Hadamard state;
- positivity, a particle Hilbert space, scattering, or unitarity;
- survival or removal of either residual deformation class;
- a quantum Cartan identity or residual moment-map algebra.

Nor does the pure-Weyl coefficient vector define a relative Einstein–Maxwell/Weyl–Maxwell anomaly. On the compact magnetic product the complete noncyclic observable restriction exists, but an action-compatible cyclic pushforward is exactly obstructed on every generic axial and polar physical fibre: the Einstein action form has inertia  $(2, 0)$ , whereas the Weyl form restricted to the complete  $q$ -primary image has inertia  $(1, 1)$ . The canonical 316-row full-cone cotangent completion remains unary cyclic for its adjoined pairing, but that pairing is not the transport of the two action pairings. Thus no target-valued relative coefficient subtraction follows from this paper. A separate exact reduced classification now shows the price of a changed theory: rank one is the minimal pairing or quadratic-Hessian deformation on both generic parity fibres, while an auxiliary repair requires one positive same- $q$  physical cohomology direction and its BV cotangent dual per parity. Contractible auxiliaries and exact current improvements do not repair the inertia mismatch. The pairing-only construction is not action-derived, and neither it nor the action/auxiliary alternatives has a full off-shell changed-action lift. Consequently none authorizes matched insertions, a relative coefficient, or a relative QME.

This last non-implication has also been checked orbit by orbit. The reduced cyclic maps pass their finite-carrier identities, but the pairing orbit has no all-row odd-pairing lift or density, the action orbit has no local master action or gauge-fixed operator, and the auxiliary orbit has no off-shell field complex, measure, or anomaly insertion. No orbit therefore carries a common source/target renormalized insertion functor. The relative class, coefficient, and even their invariance under changing repair orbit remain undefined. This is a terminal statement about the present repository data, not a no-go against constructing one explicitly modified off-shell theory in the future.

Residual transfer is specifically forbidden at present. The frozen classical contraction belongs to the strict theory and has no compensator rows. The local Wess–Zumino contribution to  $Q_1$  is fixed, but the proposition proves that it is not a unique complete renormalized operator. The anomaly-induced Paneitz/Riegert representative fixes one nonlocal solution, and the logarithmic

conformal-sector coefficient is covariant and universal through curvature order two. The selected logarithmic carrier now also has the exact Fradkin–Vilkovisky Weyl-orbit completion above. The relative strict-metric raw-to- $\square R = 0$   $R^2$  scheme shift is now exact. The FV anomaly action and structural dependence of its Ricci-scalar sector are also exact. The additive finite  $C^2$  constant and absolute dressed  $R(\widehat{g})^2$  normalization remain open. The zero-derivative algebraic cubic basis, the parity-even third-curvature five-carrier quotient and the five exact universal CPT source kernels are exact. The generic ghost calculation now proves that direct minimal-CPT substitution is obstructed by a nonminimal, Hodge-mixed Schur operator. The exact Endo reduction leaves finite one-, two- and three-Ricci insertion traces. The three-insertion row is now reduced at generic nonexceptional momentum to an exact eight-sector Feynman-simplex/Wick kernel and projected onto the ten-dimensional scalar-flat five-carrier quotient as exact rational simplex integrands. At  $x_1 = x_2 = x_3 = 1$ , all eleven projected coordinates are integrated exactly in terms of one Clausen master. The one-/two-insertion nonminimal architecture is now reduced exactly to five minimal vector/scalar resolvent carriers. Their pure-vector physical sum is evaluated exactly from CPT rows 1, 3 and 14. The three longitudinal  $D_W$  towers are exactly resummed into one normalized scalar Schur determinant. Its order- $-2$  correction is proved to lie in  $\mathcal{S}_3$ , and  $\text{Wres}(K)$ ,  $\text{Wres}(K^2)$ , and  $\text{Wres} \log S_L$  are exact. The round- $S^4$  reference finite rows, canonical  $\det_3$  tail, selected weighted modified determinant, and zeta-to-weighted defect  $5/3$  are complete as well. For the distinct generic weight-raised convention, the noncommuting BCH trace and local defect  $-\frac{1}{4} \text{Wres}(K^2)$  are exact and specialize to  $-1/3$  on round  $S^4$ . The smoothing theorem shows that the generic finite counterparts still require a full primed Green/spectral carrier. All eleven generic ghost  $n = 3$  functions, including the pole-four  $I_{29}$  row, are complete and are now added to the physical-Hessian carrier representative with the exact  $I_{28}$  quotient relation preserved. The pure-vector ghost  $n = 1 + n = 2$  slice is also integrated exactly: six channels are nonzero, five vanish, and no new transcendental master appears. Adding it gives the exact partial-BV five-carrier representative; exactly three longitudinal/mixed  $D_W$  carriers remain open in the ghost block. The same-gauge physical Hessian is now imported through first curvature order together with its projected algebraic curvature-squared block; the gauge-ordering commutator and round- $S^4$  split are exact. Its formally self-adjoint momentum vertex, full three-linear alpha polynomial and exact five-carrier projection are fixed. The isolated three- $\mathcal{H}_1$  simplex integral is logarithmically corner obstructed at the symmetric point. The exact equal-box mixed fixture is nonzero after adding all  $\mathcal{H}_1 \mathcal{H}_2$  endpoints, so universal algebraic cancellation is excluded, and its common Mellin minimal-subtraction scale row is fixed. The generic covariant Volterra lift is now exact, and all generic contact endpoint residues are exact 33-row five-carrier functions. Their symmetric-point incidence with the triangle is exact and nonzero, refuting algebraic  $H_2$  cancellation there. The generic corner residues and full incidence are now exact; their nonzero scale row fixes the Mellin-renormalized  $M_{14}$  disposition. The  $M_{14}$  singlet and standard- $S_3$  pair are now evaluated in that scheme, all physical six-master coordinate functions are exact, and all eleven physical triangle channels are integrated in the exact seven-function basis. Finite local normalization rows and assembly into the complete repository form factors are still required. The generic primed Green/spectral carrier and renormalized physical trace, hence the five complete repository form-factor functions and coefficients, the parity-odd derivative manifest, global Green/kernel data, and renormalized-product contribution remain open. The next gate is a compensator-inclusive classical contraction together with those missing data. The exact positive-Berger contraction does not fill this gate: its row named  $\tau$  is a temporal diffeomorphism ghost on a different background, not the Wess–Zumino scalar compensator. The separate same-background free-state and Bridge 4 programme is outside this paper. Bridge 5 has a QME disposition but still lacks the independent interaction-to-physical-branch map. No classical tangent obstruction is therefore called ghost removal or a quantum constraint.

## 8 Machine certification and reproduction

The coefficient-bearing theorem receipts are

- `quantum-weyl/local_bv/certificates/AFNO_H14_EVEN_CANONICAL_QUOTIENT.json`.
- `quantum-weyl/local_bv/certificates/AFNO_H14_ODD_CANONICAL_QUOTIENT.json`.
- `quantum-weyl/local_bv/certificates/AFNO_DIFF_MIXED_MINIMAL_BV_H14.json`.
- `quantum-weyl/local_bv/certificates/MINIMAL_BV_KOSZUL_TATE_COLLAPSE.json`.
- `quantum-weyl/local_bv/certificates/GENERAL_NONMINIMAL_GAUGE_FIXED_CONTRACTI  
ON.json`.
- `quantum-weyl/spectral/euclidean/certificates/REPOSITORY_EUCLIDEAN_ELLIPTIC_  
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- `quantum-weyl/spectral/euclidean/certificates/REPOSITORY_FULL_BV_MULTIPLICIT  
Y_LEDGER.json`.
- `quantum-weyl/spectral/euclidean/certificates/STANDARD_EUCLIDEAN_LOCAL_B4_IN  
TEGRATION_SLICE.json`.
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_OR_RICCI_FLAT_FULL_BV_OPERATOR_MEASURE_COEFFICIENT_MATCH.json`.
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- `quantum-weyl/anomalies/certificates/WESS_ZUMINO_EXTENDED_LOCAL_BV_COHOMOLOGY.  
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- `quantum-weyl/transfer/certificates/FLAT_TT_LOGARITHMIC_GAMMA1.json`.
- `quantum-weyl/transfer/certificates/CURVATURE_SQUARED_COVARIANT_LOG_GAMMA1.j  
son`.
- `quantum-weyl/transfer/certificates/FV_CONFORMIZED_C2_LOG_GAMMA1.json`.
- `quantum-weyl/transfer/certificates/FV_ANOMALY_ACTION_RICCI_SECTOR.json`.

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- quantum-weyl/transfer/certificates/FOUR\_DIMENSIONAL\_THIRD\_CURVATURE\_WEYL\_CARRIER\_MANIFEST.json.
- quantum-weyl/transfer/certificates/CPT\_UNIVERSAL\_THIRD\_CURVATURE\_KERNELS.json.
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- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_PHYSICAL\_HESSIAN\_CURVATURE\_SQUARED.json.
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- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_GHOST\_SCHUR\_SCHATTEN\_SPLIT.json.
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- quantum-weyl/spectral/euclidean/certificates/ROUND\_S4\_GHOST\_SCHUR\_FINITE\_WEIGHTED\_TRACES.json.
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- quantum-weyl/spectral/euclidean/certificates/PRODUCT\_S2\_S2\_GHOST\_SCHUR\_SPECTRAL\_CARRIER.json.
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- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_GHOST\_SCHUR\_WEIGHT\_RAISED\_ZETA\_FACTORIZATION.json.
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- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_GHOST\_N3\_FIVE\_CARRIER\_PROJECTION.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_GHOST\_N3\_BARYCENTRIC\_FACTORIZATION.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_GHOST\_N3\_POLE3\_RELATIVE\_IBP.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC\_SCALAR\_TRIANGLE\_DIFFERENTIAL\_SYSTEM.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_GHOST\_N3\_POLE3\_INTEGRATED\_FUNCTIONS.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_GHOST\_N3\_I29\_INTEGRATED\_FUNCTION.json.
- quantum-weyl/spectral/euclidean/certificates/GENERIC\_BACKGROUND\_GHOST\_N3\_SYMMETRIC\_POINT\_SIMPLEX\_INTEGRATION.json.
- quantum-weyl/spectral/euclidean/certificates/WEYL\_GRAVITON\_BOX\_R\_SCHEME\_CONVERSION.json.

The companion computational supplement prints the exact quotient ranks, orbit counts, coefficient vector, quartet matrices, extended boundary matrix, and input-hash ledger from generated certificate tables. The claim map binds the manuscript, both PDFs, the generated tables, their producer and verifier, and all explicit nonclaims.

The scoped replay is:

```

PYTHONPATH=quantum-weyl python3 -m \
    local_bv.verify_diff_mixed_total_complex
PYTHONPATH=quantum-weyl python3 -m \
    spectral.euclidean.verify_standard_integration_slice
PYTHONPATH=quantum-weyl python3 -m \
    anomalies.verify_repository_regulated_slavnov_breaking
PYTHONPATH=quantum-weyl python3 -m \
    anomalies.verify_unitary_matter_cancellation_no_go
PYTHONPATH=quantum-weyl python3 -m \
    anomalies.verify_wess_zumino_minimal_bv_cotangent_lift
PYTHONPATH=quantum-weyl python3 -m \
    anomalies.verify_wess_zumino_extended_local_bv_cohomology
PYTHONPATH=quantum-weyl python3 -m \
    transfer.verify_one_loop_slavnov_q1_disposition
PYTHONPATH=quantum-weyl python3 -m \
    transfer.verify_flat_tt_logarithmic_gamma1
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PYTHONPATH=quantum-weyl python3 -m \
    transfer.verify_fv_anomaly_action_ricci_sector
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PYTHONPATH=quantum-weyl python3 -m \
    transfer.verify_third_curvature_weyl_manifest
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PYTHONPATH=quantum-weyl python3 -m \
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PYTHONPATH=quantum-weyl python3 -m \
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zeta_factorization
PYTHONPATH=quantum-weyl python3 -m \
    spectral.euclidean.verify_generic_background_ghost_n3_adiabatic_carrier
PYTHONPATH=quantum-weyl python3 -m \

```



```

spectral.euclidean.verify_generic_background_ghost_n3_triangle_kernel
PYTHONPATH=quantum-weyl python3 -m \
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spectral.euclidean.verify_generic_background_ghost_n3_barycentric_factorization
PYTHONPATH=quantum-weyl python3 -m \
spectral.euclidean.verify_generic_background_ghost_n3_pole3_relative_ibp
MOD="spectral.euclidean.verify_generic_scalar_triangle_"
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transfer.verify_scalar_flat_k_ricci_crosswalk
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quantum-weyl/spectral/euclidean/verify_generic_background_\
physical_hessian_n3_integration_obstruction.py
PYTHONPATH=quantum-weyl python3 -m \
spectral.euclidean.verify_box_r_scheme_conversion
PYTHONPATH=quantum-weyl python3 -m verify_active_frontier
PYTHONPATH=quantum-weyl python3 -m atlas.verify_quantum_atlas_fragment
python3 paper/verify_12_pure_weyl_one_loop_bv_anomaly_claim_map.py

```

## 9 Outlook

The immediate local-to-residual question is now sharp. The pole-four  $I_{29}$  row and all ten pole-three ghost rows are already reduced to the certified triangle/log master system. The exact physical three-linear numerator and its five-carrier projection are now operational, as is an exact polarized mixed fixture with a fixed Mellin subtraction and scale row. The generic covariant Volterra carrier and every generic contact endpoint kernel are now fixed. The generic triangle corner residues and their full contact incidence are also exact and refute  $H_2$  cancellation. The 33 minimally-subtracted finite contact rows are now exact as well. The physical triangle's rank-52 six-master carrier span is exact, and its renormalized  $M_{14}$  singlet and standard- $S_3$  pair are now explicit sector-decomposed rational/logarithmic functions in the common Mellin scheme. The sixty-six reduced rational master-coordinate functions are now exact, and all eleven relative-IBP boundary fluxes and seven-function triangle decompositions are complete. Those functions, the finite contact rows and the combined scale rows are now assembled into five carrier-labelled physical-Hessian Mellin-MS functions, and the complete integrated ghost- $n = 3$  triangle is added in the same ten-dimensional quotient. The pure-vector ghost  $n = 1 + n = 2$  slice is integrated in the same triangle/two-log/rational basis and has been added as well. Keeping the independent finite-counterterm normalization explicit, analytically continue the exact  $S^2(k_1) \times S^2(k_2)$  bivariate Schur sums with the matched exceptional factor  $3^{-6}$  and add the remaining BV sectors. Arbitrary-background finite Schur rows still require a generic primed Green kernel or complete spectral measure. Then complete the substitutions needed to match the five imported universal CPT kernels to repository parity-even third- curvature form-factor functions

and coefficients on the certified carrier quotient, and classify the parity-odd derivative sector; then supply finite  $C^2$  and absolute dressed  $R(\hat{g})^2$  normalization conditions and renormalized BV operator data to fix a complete  $Q_1$ ; then, given an extended contraction  $(\iota_{\text{ext}}, \pi_{\text{ext}}, S_{\text{ext}})$ , compute

$$q_1 = \pi_{\text{ext}} Q_1 \iota_{\text{ext}}$$

and test first-order nilpotency, the quantum Cartan defect, incoming  $H^3 \rightarrow H^4$ , outgoing  $H^4 \rightarrow H^5$ , operator mixing, and the corrected pairing. This is a new theorem, not a corollary of local QME restoration.

The independent Lorentzian rail must construct renormalized causal products and a compatible state before any unitary or particle conclusion is possible. The present result nevertheless fixes the finite local obstruction vector that every such construction must reproduce or trivialize.

## References

- [1] I. A. Batalin and G. A. Vilkovisky, *Gauge algebra and quantization*, Phys. Lett. B **102** (1981), 27–31.
- [2] M. Henneaux and C. Teitelboim, *Quantization of Gauge Systems*, Princeton University Press, 1992.
- [3] G. Barnich, F. Brandt, and M. Henneaux, *Local BRST cohomology in the antifield formalism. I. General theorems*, Commun. Math. Phys. **174** (1995), 57–91, [doi:10.1007/BF02099464](https://doi.org/10.1007/BF02099464).
- [4] M.-F. Ouedraogo and S. Paycha, *The multiplicative anomaly for determinants revisited; locality*, Commun. Math. Phys. **273** (2007), 133–167, [doi:10.1007/s00220-007-0235-4](https://doi.org/10.1007/s00220-007-0235-4).
- [5] N. Boulanger, *General solutions of the Wess–Zumino consistency condition for the Weyl anomalies*, J. High Energy Phys. **07** (2007), 069, [doi:10.1088/1126-6708/2007/07/069](https://doi.org/10.1088/1126-6708/2007/07/069).
- [6] E. S. Fradkin and A. A. Tseytlin, *Conformal anomaly in Weyl theory and anomaly free superconformal theories*, Phys. Lett. B **134** (1984), 187–193, [doi:10.1016/0370-2693\(84\)90668-3](https://doi.org/10.1016/0370-2693(84)90668-3).
- [7] F. Baume and B. Keren-Zur, *The dilaton Wess–Zumino action in higher dimensions*, J. High Energy Phys. **11** (2013), 102, [doi:10.1007/JHEP11\(2013\)102](https://doi.org/10.1007/JHEP11(2013)102).
- [8] A. O. Barvinsky, G. H. S. Camargo, A. E. Kalugin, N. Ohta, and I. L. Shapiro, *Local term in the anomaly-induced action of Weyl quantum gravity*, Phys. Rev. D **108** (2023), 086018, [doi:10.1103/PhysRevD.108.086018](https://doi.org/10.1103/PhysRevD.108.086018).
- [9] N. Ohta and R. Percacci, *Ultraviolet fixed points in conformal gravity and general quadratic theories*, Class. Quantum Grav. **33** (2016), 035001, [doi:10.1088/0264-9381/33/3/035001](https://doi.org/10.1088/0264-9381/33/3/035001).
- [10] R. J. Riegert, *A nonlocal action for the trace anomaly*, Phys. Lett. B **134** (1984), 56–60.
- [11] S. M. Paneitz, *A quartic conformally covariant differential operator for arbitrary pseudo-Riemannian manifolds*, SIGMA **4** (2008), 036, [doi:10.3842/SIGMA.2008.036](https://doi.org/10.3842/SIGMA.2008.036) (1983 manuscript).
- [12] F. Bugini and D. E. Díaz, *Holographic Weyl anomaly for GJMS operators: one Laplacian to rule them all*, J. High Energy Phys. **02** (2019), 188, [doi:10.1007/JHEP02\(2019\)188](https://doi.org/10.1007/JHEP02(2019)188).

- [13] R. Aros, C. Núñez-Barra, F. Bugini, and D. E. Díaz, *Heat coefficients, the multiplicative anomaly, and four dimensional Casimir energy for conformally covariant powers of the Laplacian*, J. Math. Phys. **66** (2025), 062303, [doi:10.1063/5.0252371](https://doi.org/10.1063/5.0252371).
- [14] S. M. Kuzenko and M. Ponds, *Conformal geometry and (super)conformal higher-spin gauge theories*, J. High Energy Phys. **05** (2019), 113, [arXiv:1902.08010](https://arxiv.org/abs/1902.08010).
- [15] S. M. Kuzenko and M. Ponds, *Generalised conformal higher-spin fields in curved backgrounds*, J. High Energy Phys. **04** (2020), 021, [arXiv:1912.00652](https://arxiv.org/abs/1912.00652).
- [16] S. M. Kuzenko, M. Ponds, and E. S. N. Raptakis, *New locally (super)conformal gauge models in Bach-flat backgrounds*, J. High Energy Phys. **08** (2020), 068, [arXiv:2005.08657](https://arxiv.org/abs/2005.08657).
- [17] M. Beccaria and A. A. Tseytlin, *On induced action for conformal higher spins in curved background*, Nucl. Phys. B **919** (2017), 359–383, [arXiv:1702.00222](https://arxiv.org/abs/1702.00222).
- [18] A. O. Barvinsky and W. Wachowski, *Notes on conformal anomaly, nonlocal effective action and the metamorphosis of the running scale*, Phys. Rev. D **108** (2023), 045014, [doi:10.1103/PhysRevD.108.045014](https://doi.org/10.1103/PhysRevD.108.045014).
- [19] A. O. Barvinsky, Yu. V. Gusev, V. V. Zhytnikov, and G. A. Vilkovisky, *Covariant perturbation theory (IV). Third order in the curvature*, [arXiv:0911.1168](https://arxiv.org/abs/0911.1168).
- [20] A. O. Barvinsky, A. E. Kalugin, and W. Wachowski, *Schwinger–DeWitt expansion for the heat kernel of nonminimal operators in causal theories*, [arXiv:2508.06439v2](https://arxiv.org/abs/2508.06439v2).
- [21] A. O. Barvinsky and G. A. Vilkovisky, *Conformal decomposition of the effective action and covariant curvature expansion*, [arXiv:gr-qc/9510037](https://arxiv.org/abs/gr-qc/9510037).
- [22] J. F. Donoghue and B. K. El-Menoufi, *Covariant non-local action for massless QED and the curvature expansion*, J. High Energy Phys. **10** (2015), 044, [doi:10.1007/JHEP10\(2015\)044](https://doi.org/10.1007/JHEP10(2015)044).
- [23] F. Gesztesy, Y. Latushkin, and K. Zumbrun, *Derivatives of (modified) Fredholm determinants and stability of standing and traveling waves*, J. Math. Pures Appl. **90** (2008), 160–200, [arXiv:0802.1665](https://arxiv.org/abs/0802.1665).
- [24] J. Delgado and M. Ruzhansky, *Schatten classes on compact manifolds: kernel conditions*, J. Funct. Anal. **267** (2014), 772–798, [arXiv:1403.6158](https://arxiv.org/abs/1403.6158).
- [25] T. Ackermann, *A note on the Wodzicki residue*, [arXiv:funct-an/9506006](https://arxiv.org/abs/funct-an/9506006).
- [26] D. V. Vassilevich, *Heat kernel expansion: user’s manual*, Phys. Rep. **388** (2003), 279–360, [arXiv:hep-th/0306138](https://arxiv.org/abs/hep-th/0306138).
- [27] S. Paycha, *Weighted trace cochains; a geometric setup for anomalies*, [arXiv:math-ph/0503033](https://arxiv.org/abs/math-ph/0503033).
- [28] M. Kontsevich and S. Vishik, *Determinants of elliptic pseudo-differential operators*, [arXiv:hep-th/9404046](https://arxiv.org/abs/hep-th/9404046).
- [29] M. Spreafico and S. Zerbini, *Spectral analysis and zeta determinant on the deformed spheres*, [arXiv:math-ph/0610046](https://arxiv.org/abs/math-ph/0610046).